Supplementary Material

Temporal Referent Integrity: An Evaluation Diagnostic for Resolution Timing in Tool-Using Agents

Anonymous Submission

This supplement reports protocols, implementation details, evidence chronology, and extended results. Literal prompts, runner interfaces, inventories, raw outputs, and executable reports are in the separately uploaded anonymous Code and Data Supplement.

1 Benchmark and Frozen Protocol

The Matched Timing Diagnostic evaluation (internal inventory ID: TRI-v3) contains 160 tasks in 20 frozen language-template clusters. Tasks are balanced across anchored and dynamic references and across flip, stable, remove, invalidate, and name-collision transitions. The transfer evaluation contains 80 tasks over four unseen schemas (projects, expenses, inventory, and cloud deployments). The Action-Validity Stress Test (internal inventory ID: TRI-v4) contains 40 conditional tasks and distinguishes action-validity and selector-match guards. The model-facing SQLite trajectory experiment uses a frozen 40-task subset and records every query, refresh, target proposal, mutation attempt, and final database diff.

The task generator evaluates the selector in S_0, applies a controlled transition to obtain S_1, and computes the target from the instruction-conditioned resolution status. Each task-specified resolver maps a state to one entity or $\perp$; the frozen generators construct a unique eligible winner, and unresolved ranking ties are outside the inventory. At the refresh boundary, Preserve induces a bound commitment $B(q(S_0))$, whereas Reevaluate retains an unresolved query $U(q)$ for evaluation in S_1. This semantic status need not be serialized as an explicit field. Paired items share their schema, entities, transition, and action. Their instruction phrasing encodes whether the target is committed before refresh or selected afterward. Every controller receives the instruction's natural-language selector. Gold targets, generator-normalized selector fields, and pre- and post-refresh winner IDs are not passed to the model.

1.1 Evidence chronology and status

The manuscript distinguishes the original primary comparison from later replication and audit analyses. The chronology below prevents post-primary reanalyses from being read as preregistered primary estimands.

Stage	Frozen status	Role in the manuscript
Matched Timing Diagnostic	Primary/frozen before calls	Package-level Qwen Generic vs. Lifecycle-Gated estimand; GLM replication
Matched Timing Diagnostic component addenda	Post-primary; frozen before their own calls	Free actor, validity gate, mode-only, untyped-plan, and exact Compile-then-act (CTA) audits
Cross-Schema Controlled Replication	Post-primary replication; frozen before calls	New schemas and states; tests directional substitution
Matched full-history baselines	Post-primary strong baseline; frozen before own calls	Ordinary and final-step-aware history across three model families
Human agreement labels	Post-primary descriptive construct audit	Three independent labels for each of 100 original/rewrite items
Six-form human follow-up	Post-primary audit; frozen eligibility gate failed	Descriptive recruitment, instrument, and gold-boundary evidence; no fixed-rater endpoint
Human-rewrite model runs	Post-primary replication; frozen before own calls	Unchanged controllers on 50 volunteer rewrites
Full-diagnostic matched-call confirmation	Post-primary; frozen before own calls; complete	Equal calls and base actor payloads on all 160 diagnostic rows
Human-rewrite matched-call confirmation	Post-primary; frozen before own calls; complete	Equal-call transfer on 50 existing volunteer rewrites; model-dependent result
Source-derived matched-call contrast	Post-primary; frozen before own calls; complete	30 changed pairs across three source-derived substrates and model families
40-task model-facing SQLite	Secondary/frozen execution test; GLM later replication	Model-issued mutations and final database states
Cross-Schema Controlled Replication deterministic replay	Post-primary, zero API	Consequence verification over frozen target outputs, not a new behavioral replication
Pair accuracy (PairAcc) and identifiability	Post-primary, zero API	Matched, marginal, and shared-eligible reanalyses of frozen outputs
Crossed-dependence sensitivity	Post-primary, zero API	Domain, template, and two-way resampling of the primary package contrast
Call/information-matched ablation	Post-primary; both models complete	Shared compiler call and matched actor payloads; explicit-decision and enforcement contrasts
Selection regret	Post-primary, zero API	Proxy maximizers and changed-PairAcc regret in frozen candidate sets
Model-authored linguistic audit	Post-primary; transport-repaired	24 generated pairs; no complete pair is accepted by both model judges
Strengthened deterministic rule	Post-hoc	Benchmark-aware deterministic audit after first-rule failure inspection
Public-suite audits	Post-primary/descriptive	Coverage checklist for audited ToolSandbox, AppWorld, and τ^3-Bench versions
Injected audit sensitivity	Post-primary, zero API	Known-positive and one-feature-missing controls; implementation check only
Model-assisted recall triage	Post-primary, zero API	Candidate-label queue over six public suites; not independent recall calibration
Source-anchored external transfer	Post-primary; frozen before calls	Author-adapted matched contrast on pinned STATE-Bench and AgentDojo states and source tools

Table 1: Evidence status used throughout the paper. Planned or unverified analyses are not reported as results.

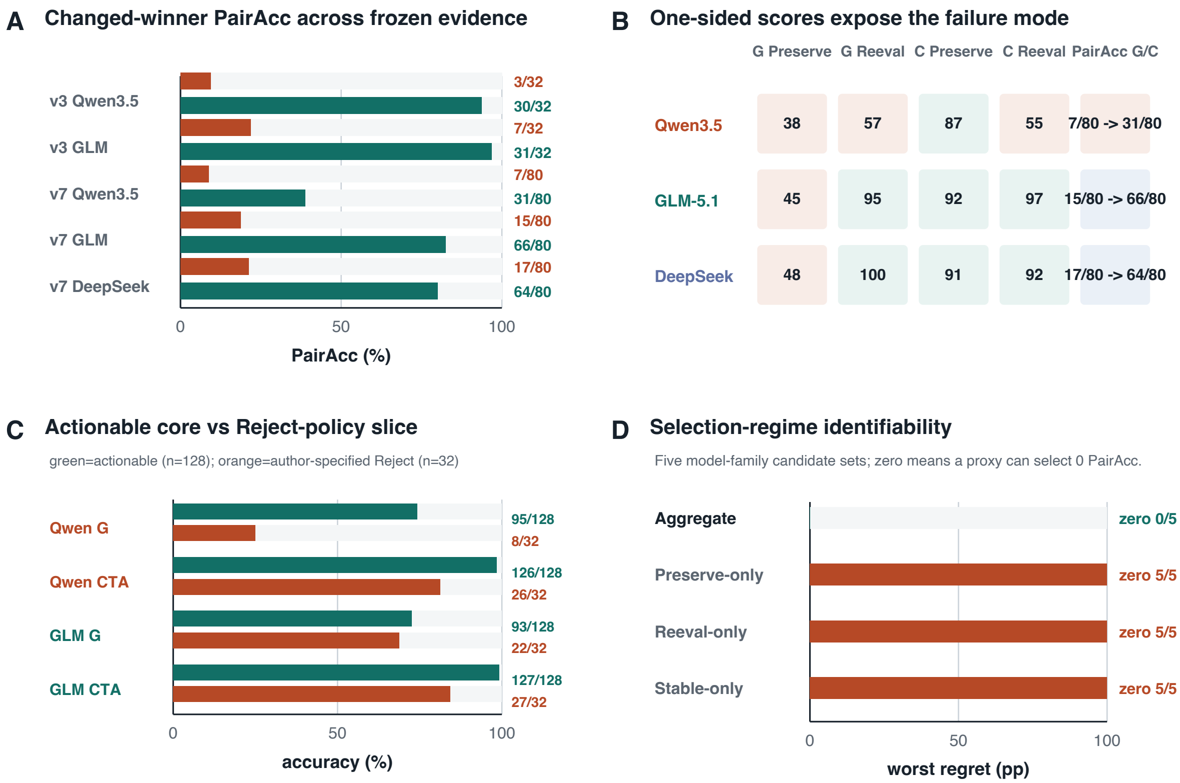

Figure 1: Post-primary audits from frozen Matched Timing Diagnostic and Cross-Schema Controlled Replication outputs. **(A)** Changed-winner PairAcc for Generic and historical CTA on 32 v3 and 80 v7 pairs. **(B)** V7 one-sided mode accuracies and complete-pair accuracy (G/C: Generic/CTA). **(C)** V3 actionable-core accuracy versus the author-specified Reject-policy slice; human support is weaker for Reject. **(D)** Worst changed-PairAcc selection regret across five model-family candidate sets; "zero" counts regimes whose maximizers include a zero-PairAcc policy.

1.2 Diagnostic matrix and metric roles

The actionable referential core crosses authorization with whether the selector winner changes. This separates structural sensitivity from success caused by an unconditional policy. Remove and Invalidate are retained as a separate execution-policy slice because Preserve may correctly yield REJECT rather than an entity ID.

Authorization	Transition	Gold target	Diagnostic role
Preserve	Stable	$q(S_0) = q(S_1)$	Ordinary-execution control
Reevaluate	Stable	$q(S_1) = q(S_0)$	Detect refresh overreaction
Preserve	Changed winner	$q(S_0)$	Detect temporal rebinding
Reevaluate	Changed winner	$q(S_1)$	Detect unconditional locking
Preserve	Remove/Invalidate	$\perp$ by policy	Validity/fallback compliance
Reevaluate	Remove/Invalidate	$q(S_1)$	Valid reselection control

Table 2: Complementary condition design. "Changed winner" comprises Flip and name collision; paired rows keep both states, selector, action, schema, and update fixed.

1.3 Evaluation-identifiability and risk results

On the actionable exact-target core, let $d \in \{0,1\}$ denote a stable or changed selector winner, $m \in \{P,R\}$ the authorization mode, $e_0 = q(S_0) \in E$, and $e_1 = q(S_1) \in E$. Both required targets are action-valid in S_1. Let L, R, and S denote deterministic Always-Lock, Always-Reevaluate, and Selective policies: L outputs e_0, R outputs e_1, and S outputs e_0 for $m = P$ and e_1 for $m = R$.

Proposition 1 (support-based observational equivalence). For any evaluation support $\mathcal{D}$: (i) if $d = 0$ on every row, L, R, and S have identical output vectors; (ii) if $\mathcal{D}$ contains no changed Preserve row, R and S are observationally equivalent; and (iii) if it contains no changed Reevaluate row, L and S are observationally equivalent. Conversely, one changed row of each mode distinguishes S from both extremes. Two rows are cardinality-minimal. Matching them on (S_0, S_1, q, a) controls content but is not required for the cardinality statement.

Proof. On Stable rows $e_0 = e_1$, so all three policies agree. On changed Reevaluate rows $R = S = e_1$; on changed Preserve rows $L = S = e_0$. Conversely, L fails the changed Reevaluate row and R fails the changed Preserve row, while S passes both. One row cannot instantiate both authorization modes. The result is restricted to these three deterministic policies, actionable rows, and exact-target observations; it neither identifies an internal mechanism nor proves a particular representation necessary.

Exact target accuracy measures end-to-end task success but can hide complementary errors. For N complete pairs, we therefore report

$$\text{PairAcc} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i^P = y_i^P \ \wedge \ \hat{y}_i^R = y_i^R].$$

Let $A_P = N^{-1}\sum_i Y_{Pi}$ and $A_R = N^{-1}\sum_i Y_{Ri}$ use this same set of complete pairs. Then the Fréchet bounds are

$$\max(0, A_P + A_R - 1) \le \text{PairAcc} \le \min(A_P, A_R).$$

Preserve and Reevaluate marginal accuracy expose which side fails, while PairAcc is the paired co-correctness score used here. API, parse, protocol, and missing-output rows are incorrect under intention-to-treat (ITT) scoring.

Proposition 2 (sharp aggregate certification bound). Let an M-row evaluation contain n disjoint complete actionable changed-winner pairs, and let aggregate binary correctness A use the same row-success definition as the pair endpoint. Then

$$\text{PairAcc} \ge \max\left\{0, 1 - \frac{M(1-A)}{n}\right\}.$$

The bound is sharp. Thus $A = 1 - n/M$ remains compatible with zero PairAcc: a controller can be correct on every non-pair row and exactly one member of every critical pair. With critical-row share $\rho = 2n/M$, aggregate accuracy $1 - \rho/2$ certifies no positive PairAcc.

Proof. Let $F = n(1 - \text{PairAcc})$ be the number of failed pairs and $E = M(1-A)$ the total row errors. Each failed pair contributes at least one row error, so $F \le E$ and rearrangement gives the bound. Equality is attained by placing one error in each of $\min(E, n)$ distinct pairs and any remaining errors outside those pairs. When $M = 2n$, the evaluation contains only complete pairs, $A = (A_P + A_R)/2$, and the result reduces to the lower Fréchet bound above. This is a certification result, not evidence of an observed controller-ranking reversal; the five completed candidate sets in this submission have zero aggregate-to-PairAcc selection regret.

Proposition 3 (strict TRI-write pathway factorization). On realized trajectories, let O be an actionable changed-winner Preserve opportunity with a completed refresh and surviving valid old target; B

the event that the observable pre-refresh binding equals e_0; U final-target substitution to the distinct e_1; and X execution of the unauthorized target-level state change on e_1. The chain rule gives

$$\Pr(O \cap B \cap U \cap X) = \Pr(O)\Pr(B \mid O)\Pr(U \mid O, B)\Pr(X \mid O, B, U).$$

No independence assumption is used. For positive denominators, the finite-sample identity is $n_X/M = (n_O/M)(n_B/n_O)(n_U/n_B)(n_X/n_U)$. If a denominator is zero, the strict event count is zero but the corresponding conditional rate is undefined, not zero. Conditional substitution estimates only the third factor; the controlled SQLite trajectories inform the fourth; authored balanced sampling does not estimate deployed $\Pr(O)$. Initial-grounding, tool-order, fallback, and other wrong writes are separate pathways and are not absorbed into this factorization.

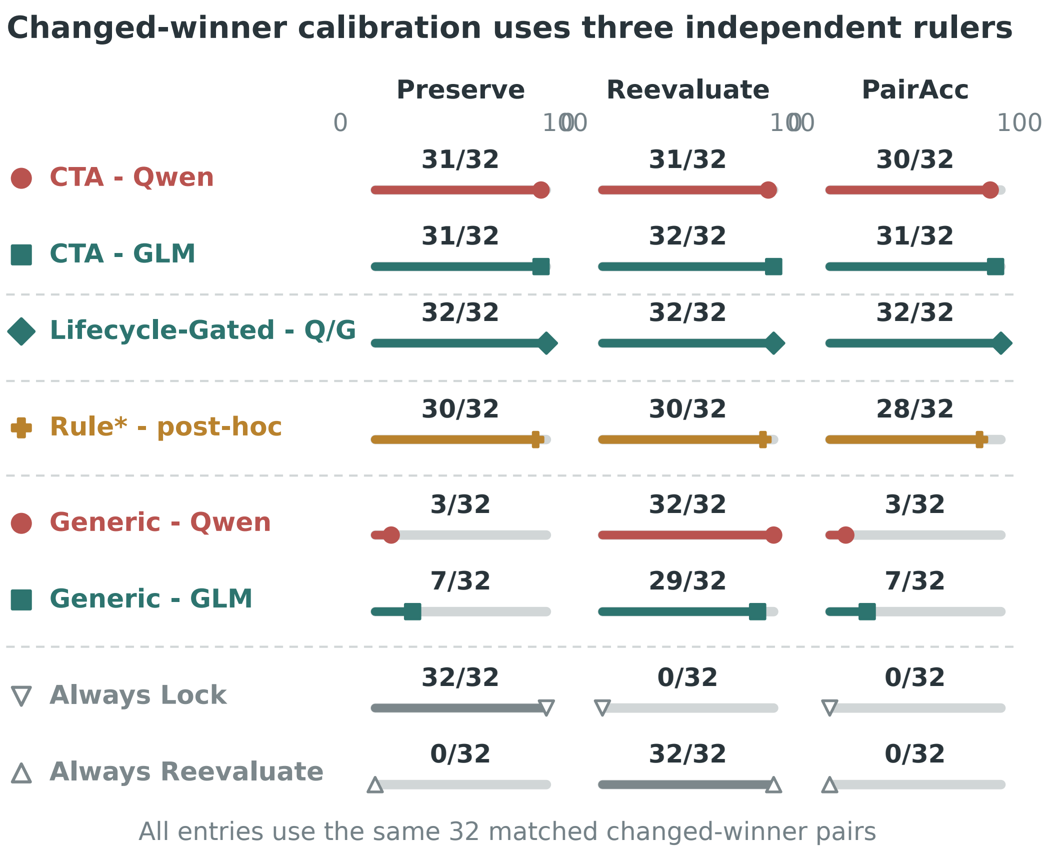

Figure 2: Changed-winner calibration on three independent rulers. Each row reports Preserve, Reevaluate, and matched PairAcc on the same 32 frozen Matched Timing Diagnostic pairs. The layout exposes complementary fixed policies without joining distinct estimands. Coincident Qwen/GLM Lifecycle-Gated results are combined; Deterministic Rule* is post-hoc.

Question	Evidence	Supported conclusion
Stable Preserve/Reevaluate semantics	Blinded labels and volunteer rewrites	Core scalar construct supported; reject fallback weaker
Aggregate-score certification	Proposition 2 and five-set selection audit	Sharp worst-case guarantee; no observed aggregate ranking reversal
Substitution after correct binding	Matched Timing Diagnostic/Cross-Schema Controlled Replication conditional audits, three models	Occurs for tested controllers under controlled conditions
Refreshed-winner direction	Flip, Stable, and name-collision controls	Supports a controller-level diagnosis, not random error
Strict write pathway	Proposition 3, conditional audit, and model-facing SQLite	One factorized pathway; no prevalence or total-risk estimate
Executed consequence	Deterministic SQLite replay	Can cause wrong-entity mutation
Dependence on CTA	CTA, Lifecycle, and post-hoc Deterministic Rule*	No uniquely necessary implementation
External-style occurrence	Mixed ToolSandbox-compatible audits	Limited and not reproduced at lower intervention
Public benchmark coverage	ToolSandbox, AppWorld, and τ^3-Bench audits	Zero strict native opportunities in audited versions
Natural prevalence	No uncontrolled traffic evidence	Not estimated

Table 3: Evidence boundary. Each row separates the completed evidence from the strongest conclusion it supports.

2 Controller Records

Executable resolution status and transition authorization are the design principle. Historical CTA is the minimal scalar pre-refresh commitment implementation and has no deterministic gate. Lifecycle adds a typed, auditable validity/fallback record; Lifecycle-Gated enforces that record at mutation time. The Generic pipeline passes the original instruction separately to its actor; its compiled ledger stores the initial selected ID, full entity snapshot, selector, requested action, and action preconditions. It omits reference mode and invalidity/fallback semantics. The scalar Lifecycle compiler additionally emits `reference_mode`, `bound_target_id`, `selector`, and `invalidity_policy`. A deterministic executor checks action validity for preserved identities and delegates only explicitly dynamic or unresolved branches to the actor. The separate Action-Validity Stress Test adds `guard_type` and `fallback_policy`; these fields are not part of the scalar Matched Timing Diagnostic record. The exact historical Compile-then-act baseline is retained unchanged as a neighboring audit.

The Action-Validity Stress Test guards are deterministic. Under `action_validity`, the controller retains the old ID iff all action preconditions $V_a(e_0, S_1)$ hold; otherwise it reevaluates $q(S_1)$. Under `selector_match`, it retains the old ID iff it is action-valid and remains $q(S_1)$; otherwise it reevaluates. The frozen 40-task inventory crosses two guard types, five instruction templates per guard, four unseen schemas, and scheduled transition types. Qwen Generic scores 21/40 (52.5%) and Lifecycle-Gated 34/40 (85.0%); this secondary extension is not part of the primary comparison. Complete task rows and the source-derived report are in the Code and Data Supplement.

2.1 Lifecycle-Gated procedure

	Lifecycle-Gated execution
1	Observe S_0 and compile $L(r, a)$ from the instruction and action schema.
2	Execute the required environment refresh and receive S_1.
3	If mode is PRESERVE, check the compiled bound ID against V_a.
4	If valid, emit the bound ID without invoking the actor.
5	If invalid, apply the compiled reject or reselection fallback.
6	For REEVALUATE or unresolved conditional branches, invoke the actor on $L(r)$ and S_1.

Table 4: Controller procedure. The instruction supplies the natural-language selector; gold, generator-normalized selector fields, and winner IDs are never provided to the compiler or actor.

2.2 Information and execution audit

Controller	Pre-refresh output	Calls	Gate
Generic ledger	ID, snapshot, selector, action, preconditions	2	no
Untyped plan	free-form plan	2	no
Historical CTA	binding time, ID, selector, reason	2	no
Lifecycle-free	mode, ID, selector, policy	2	no
Lifecycle-Gated	same typed record	1 or 2	yes

Table 5: The gated Preserve branch omits the actor call. Generic retains all task-relevant evidence and is not an information-deletion baseline.

Untyped-plan and Lifecycle-free both compile before refresh, use two calls, share the action schema and actor, and differ in whether the intermediate contract is free-form or typed. Historical CTA is not a pure representation match: its actor omits the action schema and its Preserve rule rejects only missing, rather than present-but-invalid, entities.

2.3 Verbatim Matched-Call Interface

The following system prompts are copied from the released runners. Line breaks are for typesetting only. The Generic package first compiles a ledger with:

```
Create a generic structured task-state record for a later agent step.
Record the entity selected by the target description in the current
initial state, its complete snapshot, the selector, the requested action,
and the action preconditions. Do not add specialized temporal-reference
fields such as binding_time, reference_mode, or invalidity_policy. Return
JSON only: {"task_goal":"...","selected_entity_id":"...",
"selected_entity_snapshot":{},"selector":"...","action":"...",
"action_preconditions":{}}.
```

Its final actor uses:

```
You are the final step of a tool agent after an environment refresh. Use
the original instruction, a generic structured task-state record, the
refreshed state, and the action schema to decide the target. Reason normally
about whether the old selected entity or the current selector result should
be used. If the correct target cannot be acted on, return
INVALID_BOUND_ENTITY. Return JSON only:
{"action":"process","target_id":"..."} or
{"action":"invalid","target_id":"INVALID_BOUND_ENTITY"}.
```

The full-diagnostic matched-call compiler uses:

```
Compile the instruction's target-resolution timing. Use
reference_mode="preserve" when the instruction selects or identifies the
action target before refresh and later refers to that entity. Use
reference_mode="reevaluate" when refresh happens before target selection.
For preserve, return the concrete initial selected ID as bound_target_id.
For reevaluate, return null as bound_target_id. Copy the task's selector
meaning without changing its timing. Return one JSON object only with
exactly these keys: reference_mode, bound_target_id, selector.
```

Both matched actors use the same system message:

```
Choose the exact entity ID for the requested action. Use the instruction
and supplied states exactly. The initial_selected_id is the entity selected
by the stated selector in S0. S1 is the state after refresh. If
compiler_decision is present, follow it when deciding whether to preserve
the initial entity or reevaluate the selector. If the instruction preserves
a target that is missing or violates the supplied action preconditions in
S1, return INVALID_BOUND_ENTITY rather than substituting another entity.
Return one JSON object only with exactly these keys: action, target_id. Copy
target_id exactly from a serialized state ID or use INVALID_BOUND_ENTITY.
```

Each actor payload contains `instruction`, `s0_summary`, `initial_selected_id`, `s1_state`, `selector`, `action`, and `action_schema`. Decision-visible adds one field, `compiler_decision`; removing that field produces the byte-identical History-only payload. Actor order alternates by task index. Both actors receive the original instruction, both states, the initial selection, and the same tool schema.

Table 6 gives one complete pair from the frozen inventory.

Field	Frozen value
S_0	`MTG-10`: start 540; `MTG-22`: 630; `MTG-37`: 720. All are scheduled and actionable.
S_1	`MTG-10`: start 650; `MTG-22`: 520; `MTG-37`: 720. All remain scheduled and actionable.
Selector/action	Earliest upcoming meeting; open notes; status must be scheduled and `actionable=true`.
Preserve	"Resolve the earliest upcoming meeting at this point. Following a reload of the calendar, open notes for the previously resolved meeting." Gold: `MTG-10`.
Reevaluate	"Update the calendar data, then newly identify the earliest upcoming meeting and open notes for that meeting." Gold: `MTG-22`.

Table 6: Complete actionable changed pair. States, selector, action, and schema are shared; instruction order changes the resolution time and yields opposite gold targets.

The compiler parser requires exactly `reference_mode`, `bound_target_id`, and `selector`. Preserve requires a bound ID present in a serialized state; Reevaluate requires null. The actor parser requires exactly `action` and `target_id`; the target must be a serialized state ID or `INVALID_BOUND_ENTITY`. Markdown fences may be removed before JSON parsing, but IDs are never fuzzy-matched or repaired. API, transport, parser, missing-output, and schema failures remain ITT errors. Exact-target accuracy compares the parsed ID with gold. PairAcc is one only when both pair members are exact. Conditional substitution additionally requires correct initial binding, completed refresh, a changed winner, and a surviving action-valid old target before the final ID is compared with the refreshed winner.

2.4 Executor-level oracle check

Generator-supplied representation	Accuracy	Characteristic failure
Bound ID only	41.9	premature binding
Binding time only	58.1	no anchored identity
Bound name only	58.9	alias/name collision
Latest state	64.6	temporal rebinding
Time + ID	74.0	invalidity/conditional policy
Full lifecycle schema	100.0	–

Table 7: Deterministic execution over 246 generated tasks. This establishes sufficiency for generated semantics only, not model performance, minimality, or natural-language validity.

3 Extended Component Audit

The intervention ladder is interpreted through alternatives it can rule out, not by assuming that every added component is necessary. The principal comparisons are:

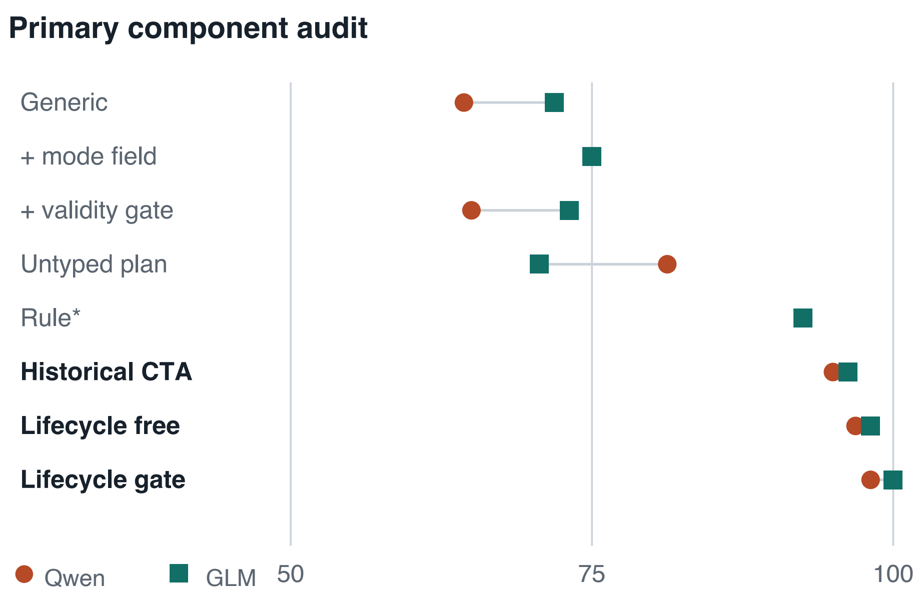

Figure 3: Component audit on Matched Timing Diagnostic exact-target accuracy. The dot-line format shows the same controller sequence for Qwen and GLM without implying that any single component is a causal estimate of the full package.

3.1 Handcrafted discourse-rule audit

The rule controller is intentionally non-neural. At prediction time it reads only the instruction, S_0, S_1, and action schema. It locates selection and refresh events in the instruction, uses event order to choose Preserve or Reevaluate, resolves the selector in the corresponding state, and applies action preconditions to a preserved ID. It cannot read the benchmark `binding`, `selector`, `update`, `style`, gold target, pre/post target, new leader, CTA record, or any model output.

Comparison	Alternative explanation tested	Result that would weaken the TRI interpretation
Generic vs. full history	Missing task evidence	Full history removes changed-winner substitution
Always-Lock vs. Always-Reevaluate	One fixed update policy suffices	Either extreme succeeds on both pair members
Stable vs. changed winner	Generic refresh instability	Comparable errors without a winner change
Validity gate vs. Generic	Action validity alone identifies the referent	Gate removes valid-target substitutions
Untyped plan vs. typed lifecycle	Any extra pre-refresh computation suffices	Call-matched plan closes the lifecycle gap
CTA/lifecycle vs. Deterministic Rule*	A learned or complex method is necessary	Deterministic discourse rule matches them
Target choice vs. SQLite replay	Errors are only representational	Substitutions do not change executed entities

Table 8: Falsification role of completed controls. A failed alternative does not prove a unique internal mechanism; it narrows the controller-level behavioral explanation.

Controller	Qwen3.5	GLM-5.1
Generic free actor	64.4	71.9
Generic + reference mode	75.0	75.0
Generic + validity gate	65.0	73.1
Untyped pre-refresh plan	81.2	70.6
Deterministic Rule* (post-hoc)	92.5	92.5
Historical CTA	95.0	96.2
Lifecycle-free actor	96.9	98.1
Lifecycle-Gated	98.1	100.0

Table 9: Exact target accuracy on the identical 160-task language set.

Rule	Timing diagnostic (160)	Human rewrites (50)	Schema replication (240)
Naive rule, frozen	60.6	74.0	77.5
Deterministic Rule*, post-hoc	92.5	96.0	91.7

Table 10: Exact authorized-target accuracy (%). The naive rule was frozen before complete-set execution. Deterministic Rule* was written after inspecting its failures across these inventories and is a strong benchmark-aware post-hoc baseline, not held-out generalization evidence.

The naive rule leaves 52/160, 12/50, and 44/240 items unresolved. Deterministic Rule* adds the observed event vocabulary and handles a single action-valid candidate without requiring a ranking field. No further changes are made after this post-hoc revision. Its frozen source SHA-256 is `34cbbd39072b...1bbfb8`; the complete digest is in the artifact manifest. On Matched Timing Diagnostic, CTA minus Deterministic Rule* is +2.5 points for Qwen and +3.8 for GLM, with both cluster intervals spanning zero. On Cross-Schema Controlled Replication, Deterministic Rule* exceeds Qwen CTA and is statistically tied with GLM and DeepSeek CTA. Thus the benchmark does not require a uniquely learned algorithm. The defensible commonality is an executable transition-authorization decision; the rule does not show that a fixed lexicon generalizes to unrestricted instructions.

After recording source SHA-256 `34cbbd39072b...1bbfb8`, we applied the unchanged rule to the 30-pair source-derived inventory frozen on STATE-Bench, AgentDojo, and ToolSandbox states and schemas. It reaches 15/60 row accuracy and 2/30 PairAcc. Table 11 keeps the source failures visible; this application is held out from Rule* development but remains a post-hoc rule evaluated on author-adapted controlled contrasts.

Source	Rows correct	PairAcc	Dominant failure
STATE-Bench	9/20	0/10	extra numeric field / wrong target
AgentDojo	0/20	0/10	unsupported timestamp selector
ToolSandbox	6/20	2/10	ranking cue absent / wrong target
All	15/60	2/30	–

Table 11: Frozen Rule* transfer to three source-derived controlled contrasts. The rule source and task inventory are hash-locked before this zero-API application. These are not native benchmark tasks or a learned-versus-rule causal comparison.

The benchmark-aware Rule* misses 20/240 Cross-Schema Controlled Replication rows, evenly split between Preserve and Reevaluate. A post-hoc zero-API residual audit evaluates frozen full-history and CTA outputs only on those rows.

Model	History-only	Timing-reminder	CTA
Qwen	10/20	13/20	13/20
GLM	10/20	20/20	20/20
DeepSeek	15/20	18/20	16/20

Table 12: Exact-target accuracy on rows missed by post-hoc Rule*. Selection uses observed Rule* errors and leaves zero complete Preserve/Reevaluate pairs, so PairAcc is undefined and these row accuracies are descriptive. The audit provides no residual CTA advantage over Timing-reminder.

3.2 Post-hoc cue and rule–model overlap audit

We use frozen text, human-majority labels, Rule* predictions, and model outputs to test whether surface triggers or Rule* solvability account for the observed pattern. This analysis is zero-API and explicitly post-hoc. A fixed trigger-count logistic is trained on v3 design labels. An order-augmented version adds only whether the first selection event precedes the first refresh event or vice versa. V3 evaluation leaves one authored template out at a time; the external diagnostic uses changed human rewrites whose determinate majority identifies either the old or refreshed target. Stable rewrites, other-entity majorities, and ties are excluded because they do not identify Preserve versus Reevaluate.

Features	v3 acc.	v3 AUC	Rewrite acc.	Rewrite AUC
Trigger counts	71.2	.724	48.4	.121
Triggers + event order	75.0	.695	48.4	.555

Table 13: Post-hoc logistic diagnostic. V3 accuracy is leave-template-out over 160 authored rows. The rewrite diagnostic contains 31 human-majority changed rows (12 Preserve, 19 Reevaluate); its majority-class baseline is 61.3%. The weak rewrite result does not replace an independent-language holdout. Existing three-annotator agreement on 100 items is $\kappa = .708$ and $\alpha = .709$.

We next cross Rule* exact correctness with each frozen model/controller on all 80 v7 changed pairs. Table 14 reports, among model-error pairs, the fraction for which Rule* gets both members correct. Most History-only and Timing-reminder errors occur on Rule*-solvable pairs. GLM CTA is the exception because it makes only 14 pair errors, ten of which coincide with a Rule* error. This overlap does not establish a model mechanism: Rule* was developed after authored error inspection and its source-derived PairAcc is only 2/30.

Model	History-only	Timing-reminder	CTA
Qwen	74.0	72.3	75.5
GLM	73.0	70.0	28.6
DeepSeek	77.9	72.3	62.5

Table 14: Percentage of each controller's erroneous changed pairs that Rule* solves. Full 2-by-2 row- and pair-level overlap counts are in the artifact report. This is a post-hoc descriptive audit, not a residual superiority test.

3.3 Concurrent Binding Drift baseline adaptation

Binding Drift (Babu and Indukuri 2026, arXiv:2607.18316) studies a primary carry slot whose gold entity remains constant across later workflow steps, together with entity-lock and large language model (LLM) re-verification defenses. We audited public commit `0e040e0954b18d4621a6f9b16f6e6e9591c822e1`. Its deterministic offline harness passes 25/25 assertions after mapping the current 200-workflow file to the legacy filename expected by the test script; the clean checkout otherwise raises `FileNotFoundError`. This path repair and the adaptation below are recorded separately.

The post-primary author adaptation applies entity-lock semantics and the public LLM self-reverification prompt frame to all 240 actionable Cross-Schema Controlled Replication rows. The verifier receives the complete temporal instruction and refreshed candidates. A short step-1 noun phrase, as used in the source benchmark, would remove TRI's timing variable. The adaptation omits S_0 and the resolved pre-refresh ID, so it is an interface audit rather than an information-matched CTA comparison or an official reproduction. The protocol was frozen before the full GLM calls.

Method	Overall	Preserve	Reevaluate	PairAcc
Entity Lock analogue	160/240	120/120	40/120	40/120
GLM self-reverify adaptation	155/240	39/120	116/120	38/120
Frozen GLM CTA	226/240	110/120	116/120	106/120
Rule* (post-hoc)	220/240	110/120	110/120	100/120

Table 15: Post-primary author adaptation of concurrent persistence and re-verification baselines. Entity Lock retains the pre-refresh ID; self-reverification resolves against refreshed candidates. The complementary mode marginals show why either unconditional policy is insufficient for matched opposite-gold evaluation.

The GLM verifier completes 240/240 rows with 240 requests, no retries or parse failures, and 89,354 recorded tokens. It substitutes the refreshed winner on 80 Preserve rows and prematurely retains the old target on two Reevaluate rows. CTA has 10 and four corresponding errors. The frozen interpretation gate classifies re-verification as a complementary-policy result. Qwen's earlier 20-task smoke selects a third, selector-ineligible visible entity on 14/20 rows, so we did not expand that grounding-confounded condition. Reports and raw outputs are `reports/binding_drift_tri_glm_v7_full_v1.*` and `runs/binding_drift_tri_glm_self_reverify_v7_full_v1.jsonl`.

Policy control	Overall	Anchored	Dynamic
Always-Lock + validity	96/160 (60.0)	80/80 (100.0)	16/80 (20.0)
Always-Reevaluate	96/160 (60.0)	16/80 (20.0)	80/80 (100.0)

Table 16: Deterministic policy extremes on the same frozen inventory. These controls are not LLM runs or reproductions of a named external agent; they expose the complementary errors of unconditional locking and unconditional selector reevaluation. The machine-readable report is `reports/policy_extreme_controls.md`.

The post-freeze Generic+reference-mode ablation adds only the explicit mode field and reaches 75.0% for both models. Its paired cluster gains over Generic are +10.6 $[+5.0, +16.9]$ for Qwen and +3.1 $[-2.5, +9.4]$

for GLM; the latter includes zero. The mode field accounts for part of CTA's gain. The audit separates three further effects. A validity-only gate changes Generic accuracy by only 0.6 and 1.2 points. Lifecycle state with a free actor reaches 96.9 and 98.1%, while the deterministic gate adds 1.2 and 1.9 points. Historical CTA reaches 95.0 and 96.2%. The call-matched untyped plan reaches 81.2 and 70.6%; Lifecycle-free exceeds it by 15.6 and 27.5 points with cluster intervals excluding zero. All high-scoring variants operationalize an executable resolution-timing decision. Typed lifecycle state provides an auditable interface, and deterministic enforcement contributes a smaller additional gain on these scalar tasks.

3.4 Matched-pair consistency

For a complete matched pair, Preserve and Reevaluate share S_0, S_1, q, a, action schema, and update. PairAcc is one only when both outcomes are correct. We pair `explicit_anchor` with `implicit_dynamic` and `implicit_anchor` with `explicit_dynamic`. Missing outputs and API, parse, or protocol failures count as incorrect under ITT; none occur in these source runs. Stable pairs are reported as controls, Flip and name collision as the changed-winner core, and remove/invalidate as a separate invalidity-policy slice. The source-derived report is `reports/matched_pair_consistency.json`.

Model	Matched Timing Diagnostic controller	All (80)	Changed (32)	Stable (16)
Qwen3.5	Generic	23 (28.8)	3 (9.4)	16 (100)
GLM-5.1	Generic	38 (47.5)	7 (21.9)	16 (100)
Qwen3.5	Historical CTA	72 (90.0)	30 (93.8)	16 (100)
GLM-5.1	Historical CTA	74 (92.5)	31 (96.9)	16 (100)
Qwen3.5	Lifecycle-free	75 (93.8)	30 (93.8)	16 (100)
GLM-5.1	Lifecycle-free	77 (96.2)	29 (90.6)	16 (100)
Qwen3.5	Lifecycle-Gated	77 (96.2)	32 (100)	16 (100)
GLM-5.1	Lifecycle-Gated	80 (100)	32 (100)	16 (100)
Independent	Always-Lock + validity	16 (20.0)	0 (0)	16 (100)
Independent	Always-Reevaluate	16 (20.0)	0 (0)	16 (100)
Independent	Deterministic Rule* (post-hoc)	68 (85.0)	28 (87.5)	16 (100)

Table 17: Matched Timing Diagnostic pairs correct (PairAcc %). The 32 invalidity-policy pairs retained in "All" are reported separately in the machine-readable table.

Model	Cross-Schema Controlled Replication controller	All (120)	Changed (80)	Stable (40)
Qwen3.5	Generic	34 (28.3)	7 (8.8)	27 (67.5)
GLM-5.1	Generic	51 (42.5)	15 (18.8)	36 (90.0)
DeepSeek-V4-Pro	Generic	57 (47.5)	17 (21.2)	40 (100)
Qwen3.5	Historical CTA	59 (49.2)	31 (38.8)	28 (70.0)
GLM-5.1	Historical CTA	106 (88.3)	66 (82.5)	40 (100)
DeepSeek-V4-Pro	Historical CTA	103 (85.8)	64 (80.0)	39 (97.5)
Qwen3.5	Lifecycle-Gated	67 (55.8)	40 (50.0)	27 (67.5)
GLM-5.1	Lifecycle-Gated	113 (94.2)	73 (91.2)	40 (100)
Independent	Always-Lock + validity	40 (33.3)	0 (0)	40 (100)
Independent	Always-Reevaluate	40 (33.3)	0 (0)	40 (100)
Independent	Deterministic Rule* (post-hoc)	100 (83.3)	60 (75.0)	40 (100)

Table 18: Cross-Schema Controlled Replication pairs correct (PairAcc %). Lifecycle-Gated was not run for DeepSeek, so no value is imputed. Low Qwen pair consistency limits any claim of universal CTA sufficiency.

3.5 Evaluation-regime identifiability audit

This post-primary, zero-API reanalysis asks which scoring regimes can identify selective authorization. "Preserve-only" and "Reevaluate-only" retain one binding mode; "Stable-only" never exercises a changed winner; "Changed PairAcc" requires both members of each matched pair. Conditional substitution additionally requires the controller's observable compiled ID to match the pre-refresh target. Failed or malformed rows are incorrect under ITT. The complete source-derived reports are `reports/v3_identifiability_regimes_v1.md` and `reports/v7_identifiability_regimes_v1.md`; no model calls were added.

On Cross-Schema Controlled Replication, Always-Lock and Always-Reevaluate both score 66.7% overall and 100% on Stable, while both obtain 0/80 changed PairAcc. Generic Reevaluate-only scores are 56.7%, 95.0%, and 100.0% for Qwen/GLM/DeepSeek, but changed PairAcc is 7/80, 15/80, and 17/80. Conditional refreshed- winner substitution is 43/72, 38/80, and 59/79. CTA removes this observed substitution (0/71, 0/70, 0/70) but retains other errors, so the conditional metric is not a general success score. These results support a bounded non-identifiability claim for the controlled diagnostics, not for all public benchmarks or natural traffic.

We then audit policy-selection consequences over these reports. The Matched Timing Diagnostic Qwen and GLM candidate sets each contain Generic, Historical CTA, Lifecycle-free, Lifecycle-Gated, Always-Lock, and Always-Reevaluate. The Cross-Schema Controlled Replication Qwen and GLM sets omit Lifecycle-free; its DeepSeek set contains Generic, Historical CTA, and the two extremes because Lifecycle-Gated was not run. An initial implementation inadvertently omitted the reported Lifecycle-Gated rows and produced a spurious 6.25-point aggregate regret. The dated protocol amendment records the omission, and the corrected report includes those rows. For each proxy, we enumerate every exact maximizer and measure regret against the best changed PairAcc in the same set.

Candidate set	Aggregate	Preserve	Reevaluate	Stable
Timing / Qwen	0.0	100.0	100.0	100.0
Timing / GLM	0.0	100.0	100.0	100.0
Schema / Qwen	0.0	50.0	50.0	50.0
Schema / GLM	0.0	91.2	91.2	91.2
Schema / DeepSeek	0.0	80.0	80.0	80.0

Table 19: Worst-case changed-PairAcc regret (percentage points) among exact proxy-score maximizers. All 15 one-sided or Stable maximizer sets include a zero-PairAcc extreme; Aggregate Aggregate end-to-end (E2E) accuracy selects a PairAcc-optimal candidate in all five complete sets. The full report lists every maximizer and optimistic regret.

Worst-case tie handling means that the proxy licenses a poor policy, not that users necessarily choose it. Maximum worst-case regret is 100 points, and the corrected audit shows no aggregate selection failure in these candidate sets. The complete post-primary, zero-API table and corrected input hashes are in `reports/evaluation_selection_regret_v1.md`.

3.6 Post-primary parser–binding–executor decomposition

We reuse frozen Matched Timing Diagnostic Lifecycle-free and Lifecycle-Gated outputs to separate three stages without new model calls. The learned condition uses the compiler's predicted mode and bound ID; oracle interventions replace only the mode, the pre-refresh bound ID, or both. The deterministic gate is then compared with the observed free actor. Reevaluate branches retain the observed actor output, so these cells diagnose component errors rather than estimate an oracle end-to-end agent.

On the primary 160-task inventory, mode accuracy and Preserve bound-ID accuracy are 160/160 and 80/80 for both Qwen and GLM. Consequently, substituting oracle mode or oracle ID does not improve Gated accuracy: Qwen remains 157/160 and GLM 160/160. Lifecycle-free is 155/160 and 157/160, respectively, isolating the small primary free-actor/gate difference after correct compilation. The unseen-schema Qwen transfer is different: mode remains 80/80, but the Preserve bound ID is only 21/40. Gated accuracy is 66/80 with learned binding and 78/80 when only the bound ID is replaced by the oracle; replacing mode alone leaves it at 66/80. Thus the transfer loss is mainly selector grounding at initial binding, not TRI or mode parsing. Complete source-derived cells are in `reports/v3_oracle_qwen_primary.json`, `v3_oracle_glm_primary.json`, and `v3_oracle_qwen_unseen.json`. This analysis is post-primary and uses oracle fields only for diagnosis, not as a deployable baseline.

3.7 Referential-core versus reject-policy sensitivity

The pre-specified aggregate includes 32 anchored remove/invalidate items whose gold execution outcome is the author-specified `INVALID_BOUND_ENTITY`. Table 20 separates these from the 128 actionable referential-core items without redefining benchmark gold or replacing the aggregate estimand. On the actionable core,

Run	Actionable core	Reject policy
Qwen Generic	95/128 (74.2)	8/32 (25.0)
Qwen Generic + validity gate	95/128 (74.2)	9/32 (28.1)
Qwen Untyped plan	106/128 (82.8)	24/32 (75.0)
Qwen Historical CTA	126/128 (98.4)	26/32 (81.2)
Qwen Lifecycle-free	123/128 (96.1)	32/32 (100.0)
Qwen Lifecycle-Gated	125/128 (97.7)	32/32 (100.0)
GLM Generic	93/128 (72.7)	22/32 (68.8)
GLM Generic + validity gate	93/128 (72.7)	24/32 (75.0)
GLM Untyped plan	107/128 (83.6)	6/32 (18.8)
GLM Historical CTA	127/128 (99.2)	27/32 (84.4)
GLM Lifecycle-free	125/128 (97.7)	32/32 (100.0)
GLM Lifecycle-Gated	128/128 (100.0)	32/32 (100.0)

Table 20: Correct/total and accuracy in parentheses. The policy slice measures compliance with the benchmark's explicit execution rule, not equally human-supported referential semantics.

Gated-minus-Generic differences are +23.4 points with a template-cluster interval of [+11.3,+38.4] for Qwen and +27.3 [+16.9,+39.8] for GLM. Gated-minus-Historical-CTA is -0.8 $[-4.9, +3.1]$ and +0.8 [0.0,+2.6], respectively; the cross-model evidence therefore does not establish Gated superiority over the strong pre-refresh predecessor. Complete reject-policy contrasts are in the artifact report.

3.8 Frozen error taxonomy

Error label	Operational meaning
Temporal rebinding	A committed pre-refresh target is replaced by the refreshed selector winner.
Premature binding	A dynamic reference incorrectly retains a pre-refresh entity.
Invalid processed	A preserved entity violates action preconditions but is still selected.
Unneeded rejection	The controller returns $\perp$ although a valid gold target exists.
Selector grounding	Mode or guard is correct, but the compiler or actor misapplies a filtered selector.

Table 21: Mechanism-oriented labels used in stage and write-consequence reports.

4 Statistical and Audit Details

The primary comparison is Lifecycle-Gated minus Generic Structured Ledger on Qwen's 20 language clusters. We resample complete clusters 10,000 times with seed 20260717. The primary Qwen difference is +33.8 points with a cluster interval of [+18.1,+50.0]; the GLM replication difference is +28.1 points with [+18.1,+38.1]. Exact paired McNemar tests are secondary: Qwen has 3 Generic-only versus 57 Lifecycle-only wins, $p = 6.25 \times 10^{-14}$; GLM has 0 versus 45, $p = 5.68 \times 10^{-14}$. API errors and missing rows are counted as failures in the audit, although the reported Matched Timing Diagnostic and Action-Validity Stress Test runs contain zero API errors and zero retries. The Matched Timing Diagnostic language-template cluster is the nearest repeated construction unit because its eight rows reuse one wording template across domains; Cross-Schema Controlled Replication analogously resamples the six rows generated from each state instance. These single-axis bootstraps do not capture every crossed dependence induced by domain, selector schema, and generator. Leave-domain and leave-template-family-out results are therefore reported as sensitivity checks below. Secondary tests and slice comparisons are not globally multiplicity-adjusted and are interpreted as exploratory rather than confirmatory evidence.

4.1 Crossed-dependence sensitivity

Because the primary inventory is a complete 8×20 crossing of domains and language-template clusters, a post-primary zero-API audit resamples each axis separately and then resamples both axes independently with multiplicative pigeonhole weights. Each method uses 10,000 draws with seed 20260725. This analysis was designed after the primary outcome; it tests sensitivity to the chosen clustering axis and does not replace the primary interval.

Model	Resampling unit	Difference (points)	95% interval
Qwen3.5	Language template	+33.8	[18.1,49.4]
Qwen3.5	Domain	+33.8	[30.6,36.9]
Qwen3.5	Two-way pigeonhole	+33.8	[16.2,50.6]
GLM-5.1	Language template	+28.1	[18.1,38.1]
GLM-5.1	Domain	+28.1	[24.4,32.5]
GLM-5.1	Two-way pigeonhole	+28.1	[16.2,40.0]

Table 22: Post-primary dependence sensitivity for Lifecycle-Gated minus Generic E2E accuracy. The widest interval for each model comes from independent resampling of both crossed axes and remains above zero. This does not address authored-language or component-attribution limits.

4.2 Full-diagnostic matched-call confirmation

This post-primary experiment was frozen before its own calls and covers all 160 rows of the Matched Timing Diagnostic. Each row makes one compiler call and two actor calls. History-only and Decision-visible receive identical base payloads, states, tool schemas, and actor prompts; only Decision-visible receives the parsed resolution-timing decision. Decision-enforced is computed offline from the visible output and makes no additional model call. All conditions are scored by intention to treat. The 32 author-specified Reject rows are reported separately and excluded from actionable substitution denominators.

Model	Outcome	Changed PairAcc	Actionable E2E	Reject	Preserve subst.
Qwen	History-only	5/32	100/128	21/32	22/28
Qwen	Decision-visible	13/32	106/128	25/32	13/28
Qwen	Decision-enforced	24/32	113/128	28/32	0/28
GLM	History-only	8/32	102/128	11/32	16/25
GLM	Decision-visible	25/32	120/128	21/32	0/25
GLM	Decision-enforced	25/32	120/128	25/32	0/25

Table 23: Post-primary full-diagnostic matched-call results. "Preserve subst." conditions on a correct observable initial binding, a surviving and actionable old target, and a distinct refreshed winner, and is restricted to the actionable core.

Decision-visible minus History-only changed PairAcc is +25.0 points [6.3,46.2] for Qwen and +53.1 [28.6,77.8] for GLM. The actionable E2E differences are +4.7 [0.0,9.4] and +14.1 [8.4,20.0]. Offline enforcement produces 18 repairs and eight harms for Qwen, and four repairs with no harms for GLM. All 960 planned logical calls complete without retry, transport, or parse failure. These results estimate decision visibility under matched calls and base payloads. The tested compilers, representations, and gates remain observationally non-unique.

4.3 Submission-critical convention and model-coverage audit

The following post-primary matrices were frozen in the submission-critical addendum. They retain authored gold and do not constitute independent-language evidence.

Model	Plain PairAcc	Convention PairAcc	Difference (95% CI)
Qwen	4/40	2/40	-5.0 [-12.5,0.0]
GLM	6/40	10/40	+10.0 [2.5,20.0]
DeepSeek	2/40	5/40	+7.5 [-5.0,20.0]
MiniMax	15/40	25/40	+25.0 [12.5,37.5]

Table 24: Convention-told natural-history control on 40 authored changed pairs. Both conditions make one actor call with byte-matched user payloads and receive no structured ID record, reference mode, compiler output, or gold. Differences are Convention minus Plain.

Model	History PairAcc	Visible PairAcc	Difference (95% CI)
Qwen	5/32	13/32	+25.0 [6.2,46.2]
GLM	8/32	25/32	+53.1 [28.6,77.8]
DeepSeek	11/32	25/32	+43.8 [16.7,71.4]
MiniMax	14/32	25/32	+34.4 [11.1,58.3]

Table 25: Four-model full-diagnostic matched-call audit on 32 actionable changed pairs. Decision-visible jointly exposes the compiler mode, bound ID, and selector restatement; the contrast is not an individual-field effect.

Source-derived repeat stability. Temperature-zero repeats use the same 30 source-derived pairs and therefore measure endpoint repeatability rather than additional independent sample size.

Model	Pass	History	Visible	Difference
Qwen	historical	12/30	13/30	3.3 pp
GLM	historical	11/30	20/30	30.0 pp
DeepSeek	historical	19/30	22/30	10.0 pp
Qwen	repeat2	10/30	15/30	16.7 pp
GLM	repeat2	12/30	19/30	23.3 pp
DeepSeek	repeat2	18/30	20/30	6.7 pp
MiniMax	first-pass	19/30	20/30	3.3 pp

Table 26: Source-derived matched-call passes. Repeated passes do not increase the 30-pair denominator; MiniMax is a first pass rather than a repeat.

ToolSandbox-style null repeat. The four versioned repeat cells preserve the original conditional denominator and keep upstream/process errors and wrong writes separate.

Model	Interface	Rows	Opportunities	Mechanism errors	Wrong writes
GLM	full history	96	73	0	14
GLM	matched generic state observed	96	86	0	3
Qwen	full history	96	67	0	6
Qwen	matched generic state observed	96	41	0	6

Table 27: Repeat of the four paper-facing ToolSandbox-style cells. These are controlled external-style trajectories, not official ToolSandbox tasks or prevalence estimates.

Decision-block stratification. A zero-API audit checks the values carried by the matched interfaces. Across all 760 authored, rewrite, source-derived, and cross-schema records, the task initial ID and every compiler or actor copy are identical (760/760). The compiler selector also equals the base selector in 760/760 records. Decision-visible therefore restates existing values; this equality does not exclude a representation or salience effect.

Model	Compiler stratum	Rows	History exact	Visible exact	Repairs / harms
Qwen	Mode correct	137	105	119	16 / 2
Qwen	Mode wrong	23	16	12	0 / 4
Qwen	Preserve ID correct	70	39	52	15 / 2
Qwen	Preserve ID wrong	10	4	2	0 / 2
GLM	Mode correct	141	110	136	27 / 1
GLM	Mode wrong	19	3	5	2 / 0
GLM	Preserve ID correct	61	32	57	25 / 0
GLM	Preserve ID wrong	19	3	5	2 / 0

Table 28: Descriptive stratification of the authored matched-call outputs. Repairs are History-only wrong and Decision-visible correct; harms are the reverse. Compiler correctness is a post-treatment stratum, so these counts do not estimate mediation or the effect of an individual field.

Figure 4: Row-level repairs and harms from the zero-call Decision-enforced transformation of Decision-visible outputs. Counts are separated for the authored diagnostic, human rewrites, and source-derived contrast. Enforcement repairs more rows than it harms for Qwen on the authored and rewrite sets, has balanced 4/4 effects for DeepSeek source transfer, and produces no harms for GLM in any setting.

4.4 Cross-schema matched-call ablation

This earlier post-primary ablation uses the 40 changed-winner pairs from Cross-Schema Controlled Replication. Each task makes one shared compiler call and two actor calls. Both actors receive the instruction, S_0, the observable initial selection, S_1, selector, action, and action schema; the Decision-visible actor alone receives the parsed compiler decision. A deterministic Decision-enforced outcome reuses the visible actor call and adds no request. The first Qwen smoke returned empty content because provider reasoning exhausted the 500-token cap. Those four failed rows are retained as infrastructure provenance. The corrected run explicitly disables thinking and changes no task, prompt, outcome, metric, or denominator.

Model	Outcome	Changed PairAcc	E2E	Preserve substitution
Qwen	History-only	12/40 (30.0)	52/80 (65.0)	16/28 (57.1)
Qwen	Decision-visible	20/40 (50.0)	59/80 (73.8)	4/28 (14.3)
Qwen	Decision-enforced	17/40 (42.5)	55/80 (68.8)	0/28 (0.0)
GLM	History-only	12/40 (30.0)	52/80 (65.0)	12/24 (50.0)
GLM	Decision-visible	24/40 (60.0)	64/80 (80.0)	0/24 (0.0)
GLM	Decision-enforced	24/40 (60.0)	64/80 (80.0)	0/24 (0.0)

Table 29: Completed equal-call comparison (post-primary). Values in parentheses are percentages. Decision-visible minus History-only changed PairAcc is +20.0 points [2.5,37.5] for Qwen and +30.0 [17.5,45.0] for GLM; substitution differences are -42.9 points $[-61.5, -24.2]$ and -50.0 $[-70.6, -30.0]$.

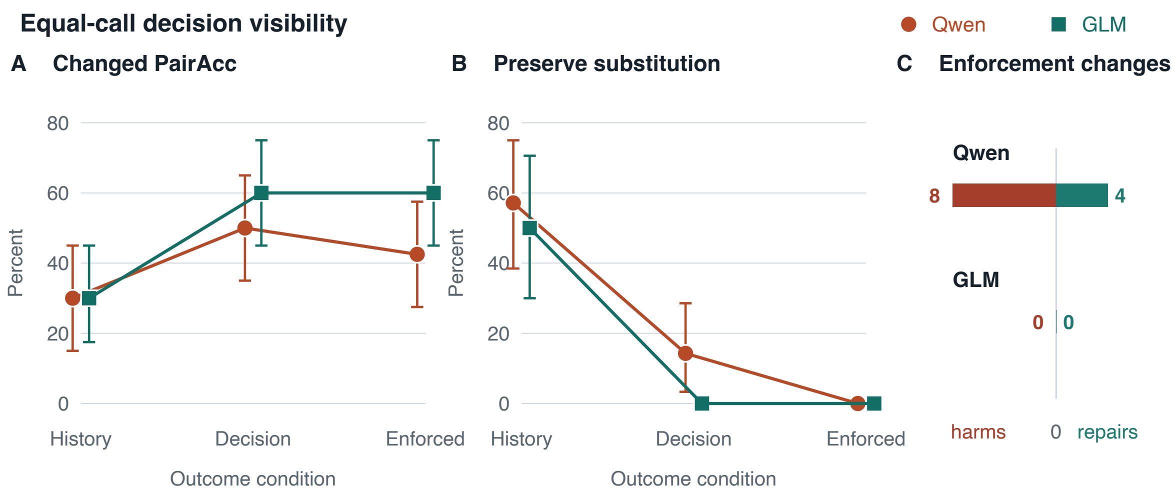

Figure 5: Outcomes with equal actor inputs and state-cluster 95% intervals. The decision block improves joint pair accuracy and reduces Preserve substitution for both models. Enforcement introduces Qwen harms and repairs but does not change GLM targets.

The compiler predicts the correct mode on 61/80 Qwen and 64/80 GLM rows; correct Preserve binding is 28/40 and 24/40. Qwen enforcement changes 16 outcomes, with four repairs and eight harms; its PairAcc contrast against Decision-visible is -7.5 points $[-22.5, 7.5]$. GLM enforcement changes no outcome. All 480 logical calls complete without retry or parse failure. This result removes differences in call count and base actor payload for the explicit-decision contrast, not every difference in compiler prompting or representation.

4.5 Cross-Schema Controlled Replication (internal inventory: TRI-v7)

The post-primary replication was frozen before its model calls with SHA-256 prefix `2504f4979f1b`; the artifact manifest records the complete hash. It contains 240 tasks spanning ten schemas not used in the primary diagnostic or transfer set. Each schema has four independently parameterized state instances, and the inventory balances binding, update, and explicitness factors. All old targets remain present and action-valid. We resample the 40 state-instance clusters.

Generic conditional substitution is 59.7% for Qwen, with a state-cluster 95% confidence interval (CI) of [46.5,72.4], and 47.5% for GLM [35.0,61.3]. CTA-minus-Generic overall differences are +23.3 points [16.2,30.4] and +24.2 [19.6,28.7]. Fixed-executor replay instantiates all 43 and 38 Generic core substitutions as wrong-entity writes. CTA/Gated produce zero core substitution writes after a correct binding, but still produce 8 and 17 Qwen wrong writes, respectively, and 14 and 7 GLM wrong writes, respectively, from opposite-mode or selector errors. Thus the replication supports the conditional phenomenon and commitment

Model	Controller	E2E target	Initial bind	Cond. subst.	Wrong writes	Stable err.
Qwen3.5	Generic	47.5	107/120	43/72	44/240	4/40
Qwen3.5	Historical CTA	70.8	103/120	0/71	8/240	6/40
Qwen3.5	Gated	71.2	95/120	0/64	17/240	9/40
GLM-5.1	Generic	70.0	120/120	38/80	38/240	2/40
GLM-5.1	Historical CTA	94.2	104/120	0/70	14/240	0/40
GLM-5.1	Gated	97.1	119/120	0/79	7/240	0/40
DeepSeek-V4-Pro	Generic	73.8	119/120	59/79	60/240	0/40
DeepSeek-V4-Pro	Historical CTA	91.2	107/120	0/70	17/240	1/40

Table 30: Frozen new-state replication with distinct denominators. E2E target uses all 240 rows; initial binding uses 120 anchored rows; conditional substitution requires a correct binding, a changed winner, and a still-valid old entity; wrong writes count all non-gold mutations; stable error uses 40 anchored Stable controls.

compilation, not general write safety. The original six Qwen/GLM 240-row runs contain zero API, parse, or internal errors. GLM CTA records one successful transport retry; the other five runs record none.

The later DeepSeek run uses the same frozen 240 tasks and inference settings. Generic substitutes on 59/79 opportunities, 74.7% with state-cluster interval [62.5,86.1], while CTA substitutes on 0/70. CTA-minus-Generic accuracy is +17.5 points [10.8,23.3]. Fixed-executor replay instantiates all 59 Generic core substitutions as wrong-entity SQLite writes. Across all errors, Generic and CTA produce 60 and 17 wrong writes; CTA's remaining errors are outside the conditional-substitution definition. Both 240-row files have zero API/parse errors and zero retries.

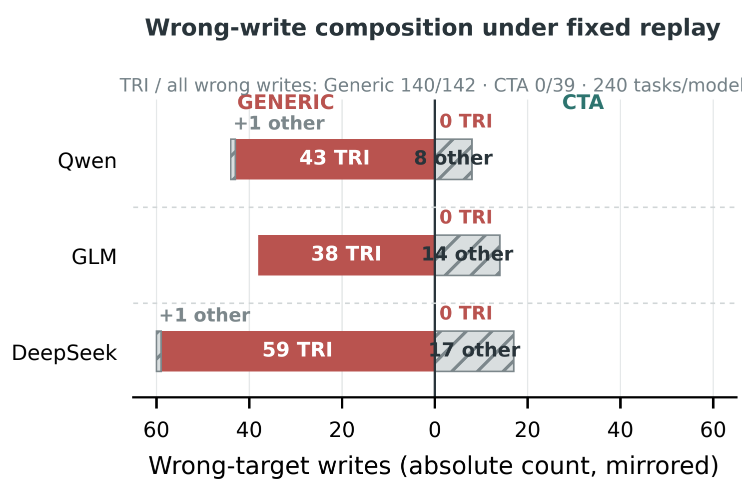

Figure 6: Fixed-executor decomposition for the 240-task controlled replication. Generic counts are mirrored on the left and CTA counts on the right. Coral segments are conditional TRI writes; hatched segments are other/non-core wrong-target writes. This deterministic replay verifies the selected-target-to-write mapping and is not a model-facing or prevalence estimate.

4.6 Matched Full-History Baselines

After the Cross-Schema Controlled Replication and DeepSeek results were known, we froze a separate strong-baseline protocol on the unchanged 240 tasks. `interactive` is an ordinary two-call trajectory: the first response requests refresh and the second retains the complete conversation, without TRI or reference-mode terminology. `full_history_once` is a strong one-call late-authorization baseline that sees both states and

Model	Shared subst. G/C	Generic PairAcc	CTA PairAcc	CTA−G
Qwen3.5	41/66; 0/66	7/80 [2.5,16.2]	31/80 [26.2,51.2]	+30.0 [16.2,43.8]
GLM-5.1	30/70; 0/70	15/80 [8.8,30.0]	66/80 [73.8,90.0]	+63.7 [52.5,75.0]
DeepSeek	50/69; 0/69	17/80 [11.2,32.5]	64/80 [70.0,88.8]	+58.8 [43.8,72.5]

Table 31: Post-primary zero-API Cross-Schema Controlled Replication audit. Shared substitution uses only tasks where Generic and CTA both expose the correct initial ID; G/C denotes Generic/CTA. PairAcc entries give counts and state-cluster bootstrap 95% intervals in percent; the final column is the paired percentage-point difference. Zero observed CTA substitutions do not establish zero population risk.

is explicitly asked whether to preserve, reevaluate, or reject. Historical CTA uses the existing matched replication run. Temperature is zero, thinking is disabled, and the output cap is 1,200.

Model	Interactive	Aware	CTA	Int. subst.	Aware subst.	CTA–Aware CI
Qwen3.5	63.3	69.6	70.8	77/80	56/80	+1.2 [-6.7,9.2]
GLM-5.1	67.1	80.8	94.2	74/80	38/80	+13.3 [8.8,17.9]
DeepSeek	68.8	75.8	91.2	68/80	42/80	+15.4 [9.2,21.7]

Table 32: Matched Cross-Schema Controlled Replication exact-target accuracy (%) and anchored changed-winner substitutions. Intervals resample 40 state clusters. For Qwen, the CTA–Aware interval includes zero.

All six files contain 240 rows and zero final API/parse failures. Qwen aware uses 269 request attempts with 29 retries; the other one-call aware runs use 240 attempts and zero retries. Interactive runs use 480 attempts. Since neither full-history condition exposes a separately scored pre-refresh selection, its substitution rate is unconditional and is not a conditional substitution estimate. SQLite replay converts the interactive errors into 87/79/75 wrong writes and the aware errors into 70/46/57 wrong writes for Qwen/GLM/DeepSeek, with four invalid attempts and one unnecessary rejection across all six runs. Thus complete history and a late semantic reminder do not remove wrong-entity consequences consistently across controllers; Qwen's statistically indistinguishable comparison means the result does not establish universal CTA superiority or pre-refresh necessity.

5 Qualitative Trace Cases

Table 33 places positive, negative, and excluded cases side by side. All rows are extracted from frozen raw outputs and executable state diffs but were selected after the complete runs for explanation. They show that the reported behaviors occurred in the tested model–tool loops; they are not a random sample or evidence of natural prevalence.

Case	User instruction (abridged only for layout)	Observable model/tool trace	Classification
Schema / Qwen	Choose the earliest incomplete reminder from the present view; reload; keep that chosen reminder as the action target.	bind `REM-1A` → target/write `REM-1B`	Post-binding substitution; controlled SQLite
TS / GLM	Before synchronizing, find the newest-created editable reminder; afterward move that reminder, without switching entries.	bind `REM-A` → `postpone(REM-B)`	Post-binding substitution; custom, post-hoc case
AW / GLM	Find the earliest task in the list as it is now; synchronize, then postpone the task found.	bind `6334` → `postpone(6334)`	Eligible opportunity; no violation
AW / Qwen	Find the alphabetically first note as the notes are now; synchronize, then append to the note found.	sync → late bind/write `3086`	Tool-order error; not TRI

Table 33: Source-validated qualitative cases. Exact instructions, raw model actions, full traces, and source paths are in `reports/qualitative_trace_cases.json`. The external rows use custom tasks on ToolSandbox-compatible (TS) or AppWorld-backed (AW) environments and are not official benchmark results.

Frozen cross-schema write trace. Task `tri-v7-core-reminders-s1-explicit_anchor-name_collision` asks Qwen3.5 Generic to choose the earliest incomplete reminder now, reload, and postpone that same reminder. In S_0, `REM-1A` (display "Renewal," due 110) wins; the compiled ledger records `selected_entity_id=REM-1A`. In S_1, `REM-1A` remains present, incomplete, and actionable, while `REM-1B` changes to due 103 and the same display "Renewal." The actor's final response is `"target_id":"REM-1B"`. Replay executes `mutate_entity(target=REM-1B)` and records `wrong_entity_write`; CTA on the same task emits and writes `REM-1A`. There are no API, parse, or protocol errors. Sources are `runs/v7_qwen_generic_structured_ledger_then_act_full.jsonl`, `runs/v7_qwen_compile_then_act_full.jsonl`, and `runs/v7_core_sqlite_replay.jsonl`.

ToolSandbox-compatible positive intervention. The post-hoc selected custom scenario ID is

`ts2-newest-created-preserve-flip`.

GLM-5.1 Generic searches `[REM-C,REM-A]`, records `selected_entity_id=REM-A`, synchronizes, then searches `[REM-C,REM-A,REM-B]`. Although `REM-A` remains editable, it calls `postpone_reminder(REM-B)` and the database diff records a wrong write. Matched Lifecycle writes `REM-A`. This uses ToolSandbox's reminder database and native tools, but it is a custom benchmark-compatible intervention, not an official score. Source: `runs/toolsandbox_tri_matched_glm_heldout_v1.jsonl`.

Opportunity without violation. In custom AppWorld task `appworld-todoist-p1-preserve-flip`, GLM binds task `6334` before synchronization; new task `6336` becomes earliest afterward, but the trace calls `postpone_task(task_id=6334)`. Thus a changed-winner opportunity need not produce a violation. Source: `runs/appworld_tri_Pro_zai-org_GLM-5.1_full_history_full_v1.jsonl`.

Excluded pre-binding write. Qwen task `appworld-simple-note-p1-preserve-flip` first searches, synchronizes, searches again, and only then records refreshed note `3086` before writing it; the intended initial target is `3084`. Because no correct pre-refresh binding exists, this is a pre-binding tool-order error, not conditional substitution. Source: `runs/appworld_tri_simple_note_Qwen_Qwen3.5-122B-A10B_full_history_full_v1.jsonl`. The source-derived compact fields and traces for all four cases are in `reports/qualitative_trace_cases.json`.

The Qwen 40-task SQLite run gives Generic Structured Ledger 27/40 final states and 13 wrong-entity writes; the GLM run gives 26/40 and 8 wrong-entity writes. A conditional audit requires a correct initial binding, an anchored flip/name-collision update, continued presence and action-validity of the old entity, and a final write to the refreshed winner. Under this strict definition, core substitution writes occur in 8/8 Qwen and 6/8 GLM opportunities, versus 0/4 stable controls for each model. The remaining 5 and 2 wrong writes follow removal or invalidation and are reported separately as reject-policy errors. On Qwen, Generic plus a validity-only gate remains 27/40, while Lifecycle-free and Lifecycle-Gated reach 40/40. The separately run GLM Lifecycle-Gated replay also reaches 40/40; GLM Lifecycle-free was not run on SQLite. All reported 40/40 cells have zero wrong writes and collateral modifications. These controlled mutation results condition on correct compilation of the lifecycle record.

6 External Boundary Check

6.1 Source-derived matched-call contrast

This post-primary protocol was frozen before its own calls. It contains 30 changed-winner pairs, ten each grounded in STATE-Bench, AgentDojo, and ToolSandbox states and tool schemas. The refresh and opposite-gold instructions are author-supplied controlled interventions over those source substrates. Each model–task record uses one shared compiler call and two actors with equal base payloads and call counts. Decision-enforced is computed offline.

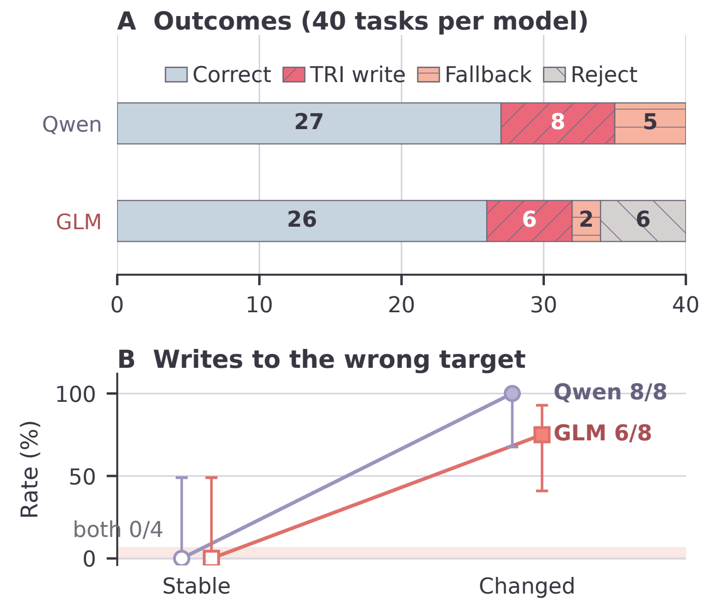

Figure 7: Outcomes for the 40 SQLite tool-loop trajectories summarized in the main paper. Panel A separates correct final states, TRI writes to the wrong target, writes in fallback cases, and unneeded rejections. Panel B compares write rates among cases meeting the strict eligibility conditions.

Model	Outcome	PairAcc	E2E	Preserve subst.	Wrong writes
Qwen	History-only	12/30	39/60	7/26	16/60
Qwen	Decision-visible	13/30	39/60	7/26	17/60
Qwen	Decision-enforced	17/30	43/60	0/26	15/60
GLM	History-only	11/30	37/60	10/30	16/60
GLM	Decision-visible	20/30	48/60	1/30	5/60
GLM	Decision-enforced	22/30	52/60	0/30	4/60
DeepSeek	History-only	19/30	45/60	6/27	11/60
DeepSeek	Decision-visible	22/30	47/60	2/27	9/60
DeepSeek	Decision-enforced	20/30	47/60	0/27	13/60

Table 34: Source-derived matched-call results. Preserve substitution conditions on correct observable initial binding, a surviving actionable old target, and a distinct refreshed winner. "Wrong writes" counts all fixed-executor wrong-target writes, not only conditional TRI errors.

Decision-visible minus History-only PairAcc is +3.3 points [−11.1,20.0] for Qwen, +30.0 [0.0,55.6] for GLM, and +10.0 [−10.0,30.0] for DeepSeek. Corresponding E2E differences are 0.0 [−6.7,6.7], +18.3 [8.3,30.0], and +3.3 [−5.0,11.7]. Thus only the GLM E2E interval excludes zero. Offline enforcement records 9/5, 4/0, and 4/4 repairs/harms for Qwen, GLM, and DeepSeek. The complete 540 logical calls have no retry, transport failure, or parse failure.

Source	Qwen	GLM	DeepSeek
STATE-Bench	8/10 → 8/10	5/10 → 10/10	10/10 → 9/10
AgentDojo	4/10 → 5/10	4/10 → 4/10	6/10 → 7/10
ToolSandbox	0/10 → 0/10	2/10 → 6/10	3/10 → 6/10

Table 35: PairAcc by source, shown as History-only → Decision-visible. Null and adverse changes are reported separately for each source.

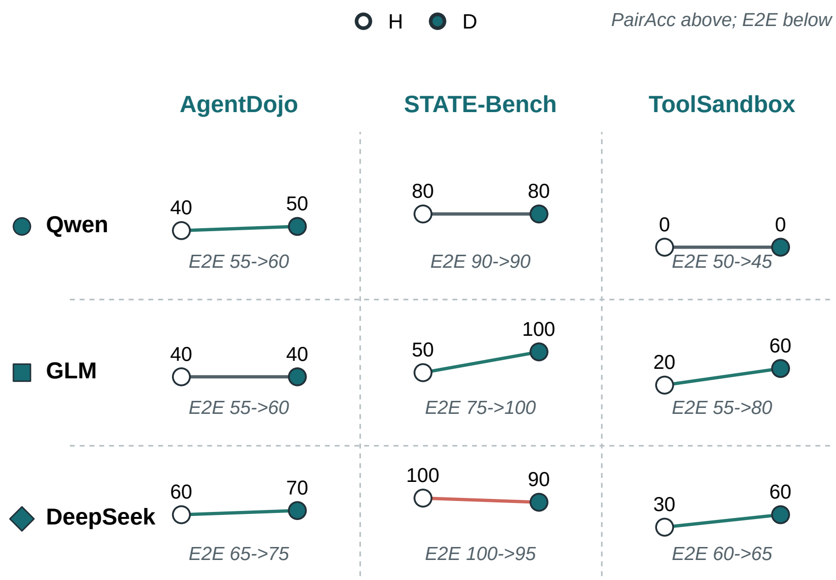

Figure 8: Source-derived controlled-contrast fingerprints. Open and filled points show History-only and Decision-visible PairAcc; the text below each cell gives actionable E2E. The source–model slices preserve the null and adverse transfers hidden by a pooled summary.

The event-order rule was hash-frozen before this application and reaches 15/60 exact targets and 2/30 PairAcc. The model-dependent actor results also vary across sources. Neither the fixed rule nor timing visibility transfers uniformly across the three source-derived substrates.

6.2 Earlier external pilots and audits

The ToolSandbox-based 24-task pilot is exploratory and is not an official benchmark score. For Generic, Lifecycle, Gated, and Untyped, respectively, accuracies are 79.2/87.5/91.7/91.7% for GLM and 91.7/83.3/91.7/79.2% for Qwen. Gated includes one GLM and two Qwen wrong writes. It therefore demonstrates that the target-integrity diagnosis can be instantiated in a stateful external-style tool loop, but does not establish a stable method advantage outside the controlled TRI setting.

Injected checklist sensitivity. We tested the deterministic public-suite audit implementation using schema-shaped controls for six suites. For each suite, five strict positives satisfy every checklist condition and five hard negatives omit exactly one required condition. After the generator and checker were frozen, the implementation retrieves all 30/30 positives and excludes all 30/30 hard negatives. This verifies the code path on known labels. It does not estimate recall for naturally phrased workflows, validate semantic candidate retrieval, or show systematic undercoverage in the public suites.

Model-assisted recall triage. A zero-API, model-assisted triage pass re-read 72 natural candidate records from ToolSandbox, AppWorld, τ^3-Bench, API-Bank, the Berkeley Function Calling Leaderboard (BFCL) Patil et al. (2026), and ToolTalk Farn and Shin (2023), together with 60 injected controls. It labeled 0/72 natural records as strict native opportunities while recovering all 30 strict-positive controls and excluding all 30 hard-negative controls. We use this only as an engineering check on candidate labeling and promptable documentation; it is not an independent recall calibration and is not treated as additional behavioral evidence.

A post-hoc strict audit joins each trace to the frozen manifest and counts only Preserve/Flip rows where the compiled initial ID is correct, the old entity remains present, and the refreshed winner is a different stable ID. GLM Generic writes the refreshed winner in 3/6 opportunities, with 0/2 wrong writes on eligible Stable controls; GLM Lifecycle is 0/5. Qwen Generic is 0/6, while Qwen Lifecycle-free is 2/6 and deterministic gate replay reduces this to 0/6. Untyped plans expose no auditable compiled ID and are excluded from this conditional denominator. The violating GLM tasks span newest-created, oldest-created, and alphabetically-last selectors. This is positive evidence that TRI can occur in a benchmark-compatible database/tool substrate, but the inventory has only six selector clusters and the strict audit was specified after the outputs existed. We therefore report task IDs and no confirmatory interval or prevalence claim.

We then ran a separate frozen 96-task single-user-turn ToolSandbox-based extension with a Preserve/Reevaluate $\times$ Stable/Flip design and an auditable initial-binding opportunity. Qwen and GLM full-history agents had zero conditional substitutions in 70 and 73 opportunities; matched Qwen and GLM Generic Ledgers had zero in 64 and 87 opportunities. The runs still produced 6, 13, 5, and 4 wrong-entity writes, respectively, but all were initial selector or tool ordering errors and therefore excluded from the post-binding numerator. This negative external result is retained as a boundary check rather than merged with the 24-task pilot or interpreted as a prevalence estimate. A separate Qwen-only `generic_state_observed` condition is retained in the artifact as exploratory sensitivity evidence (0/73 substitutions, six wrong writes, and 13 prohibited-schema/process errors). Its unmatched interface and protocol failures exclude it from these four paper-facing conditions, so it is reported as a separate sensitivity analysis.

6.3 Official opportunity and released-trace audit

We audited the unmodified public inventories at the semantic-task level. At ToolSandbox commit `165848b`, 129 semantic scenario families generate 1,032 tool-presentation instances. None contains the strict sequence correct binding, independent external transition, and later substitutable entity mutation. One multiple-user-turn task is TRI-like: after finding all friends and changing those stable person IDs to enemies, "them" refers to the same IDs when changing them back. It has no external refresh, competitor, or selector flip.

The downloaded AppWorld `0.1.3.post1` release contains 732 task instances from 244 generator families. It likewise has no independently scheduled post-binding transition, but Todoist family `8ce6779` supplies a natural action-induced near-match. The agent selects incomplete tasks assigned to the user, reassigns them, and then must "leave a comment there" on the same task IDs even though reassignment makes them fail the initial selector. We audited all 42 released trajectories for this family across 14 experiment configurations. Among 16 operations on gold task IDs, all 16 subsequent comments preserve the same ID and none substitutes another target. There are 152 omitted gold targets, two assignments to non-gold IDs, and three comments on non-gold IDs; only two trajectories pass every official test. These are upstream selection and coverage failures, not post-binding substitutions after a correct target operation.

At the current τ^3-Bench commit `cf71a80`, we further audit 2,449 core tasks: 50 airline, 114 retail, and 2,285 telecom. Airline and retail expose no user-side mutation during conversation. Telecom has 2,250 tasks with user mutation actions, but none carries a customer, line, or bill stable ID, and its single-device/app actions create no competing same-role selector target. The strict native TRI count is therefore zero. Eight telecom tasks are natural dual-control near-matches: the Agent binds an overdue bill, the user pays it, and the Agent resumes an associated line. Bill and line are different referential roles, so these are not TRI. The 26 released result files contain 10,832 trajectories from GPT-4.1, GPT-4.1-mini, Claude 3.7, and o4-mini; 936 trajectory IDs match the payment family, but none supplies a strict TRI denominator.

Strict-opportunity feature	ToolSandbox	AppWorld	τ^3-Bench
Stable entity ID	yes	yes	no
Observable prior binding	yes	yes	partial
Independent post-binding transition	no	no	yes
Competing same-role entity	no	partial	no
Changed selector winner	no	partial	no
Old entity remains actionable	yes	yes	no
Later target mutation	yes	yes	no
Evaluable authorized target	yes	yes	partial

Table 36: Coverage checklist for each public benchmark's closest natural near-match, not all tasks. "Partial" marks a related but non-strict structure. Every cell is justified in `reports/benchmark_coverage_checklist.json`. No suite contains all required features.

The missing conjunction means that these benchmarks may be unable to measure TRI; it does not

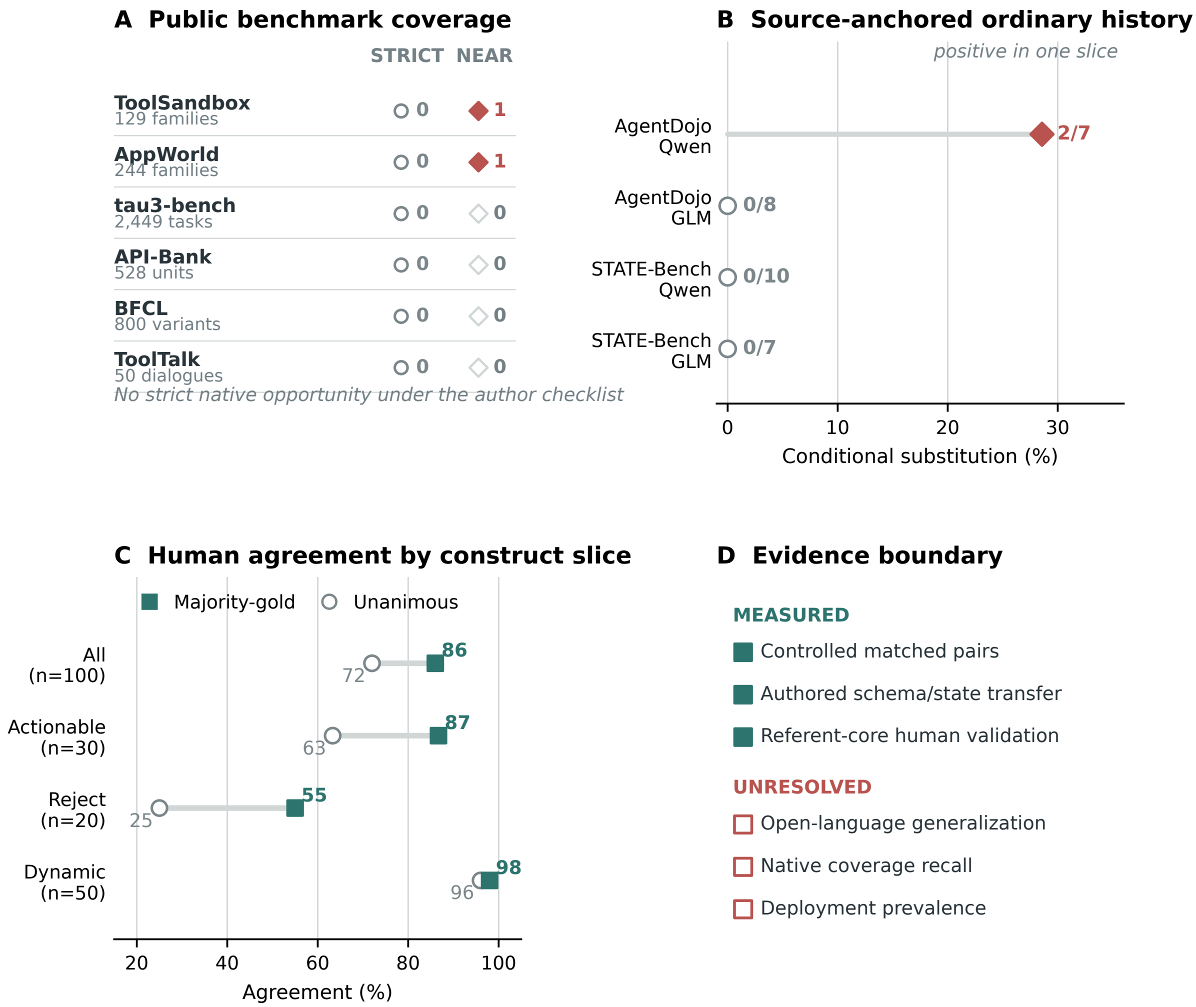

Figure 9: External validity boundary and human evidence slices. **(A)** Six public-suite audits find zero strict native opportunities under the author checklist. **(B)** Ordinary-history source-anchored transfer is positive only for Qwen on AgentDojo (2/7). **(C)** Human agreement supports the referent core more strongly than the Reject policy. **(D)** Controlled diagnostic evidence is available; open-language generalization, native coverage recall, and deployment prevalence remain unresolved.

show that TRI is frequent in deployed traffic. Conversely, a near-match is evidence of task structure, not of model violation. The source-derived funnel makes attrition explicit: ToolSandbox narrows from 129 families to 43 after entity-mutation filtering, then 26/18/1/0 after transition, prior-selection, stable-ID, and strict-opportunity checks. AppWorld narrows from 244 families to one near-match and zero strict exogenous opportunities; its 42 released trajectories expose 16 auditable post-binding operations and zero substitutions. In τ^3-Bench, 2,250/2,449 tasks contain a user-side mutation but none exposes a stable ID in that mutation. These are sequential author classifications, not an independent recall audit or a marginal feature-prevalence estimate. Full counts are generated in `reports/public_suite_coverage_funnel_v1.json`.

A separate post-primary, zero-API structural scan parses three additional pinned test inventories: API-Bank (528 units), BFCL multi-turn (800 variants in 200 clusters), and ToolTalk (50 dialogues). Author-built retrieval retains 0, 57, and 23 source-anchored candidate clusters, respectively, but finds zero strict native opportunities in each corpus. No released unit jointly exposes an observable pre-update binding, independent refresh, surviving action-valid old target, distinct refreshed winner, timing authorization, and target-level mutation. These are counts under the retrieval procedure, without independent recall calibration or prevalence meaning. The 80 BFCL/ToolTalk candidates support only further source inspection, not a

Source	Model	Condition	Exact ITT	Initial	P/S	P/C	R/S	R/C	P/C→new	R/C→old
AgentDojo	GLM	Record	33/40	31/40	8/8	8/8	8/8	7/7	0/8	0/7
AgentDojo	GLM	History	33/40	32/40	8/8	8/8	8/8	8/8	0/8	0/8
AgentDojo	Qwen	Record	27/40	25/40	6/6	7/7	6/6	4/6	0/7	2/6
AgentDojo	Qwen	History	28/40	28/40	7/7	5/7	6/7	6/7	2/7	1/7
STATE-Bench	GLM	Record	24/40	28/40	5/7	5/7	5/7	6/7	0/7	0/7
STATE-Bench	GLM	History	31/40	28/40	6/7	7/7	6/7	7/7	0/7	0/7
STATE-Bench	Qwen	Record	40/40	40/40	10/10	10/10	10/10	10/10	0/10	0/10
STATE-Bench	Qwen	History	40/40	40/40	10/10	10/10	10/10	10/10	0/10	0/10

Table 37: Frozen source-anchored external transfer. P/R denote Preserve/Reevaluate; S/C denote Stable/Changed. Four cell columns condition on a correct observable initial selection and a surviving action-valid old target. The final columns isolate substitution to the refreshed winner in Preserve/Changed and retention of the initial winner in Reevaluate/Changed. ITT retains parse and missing-write failures.

native TRI positive or behavioral transfer result; the protocol, manifest, inventory, parser, and report are included in the Code and Data Supplement.

A frozen post-primary annotation then sends each of these 80 candidates to Qwen3.5 and GLM-5.1. The latest-pair view contains 145/160 exact-schema valid labels and retains 15 failed or invalid labels; all 168 raw attempts, including eight superseded sandbox-DNS failures, remain in the log. Among 65 candidates with two valid labels, 24 have a model disagreement. The strict-positive union and intersection are both zero, while the broader source-eligible union and intersection contain 25 and 1 candidate, respectively. Thus no strict-positive source verification is triggered. These are fallible model candidate labels, not independent review, recall calibration, a verified native opportunity, a behavioral result, or a prevalence estimate.

6.4 Frozen source-anchored external transfer

We separately constructed a matched behavioral transfer from pinned STATE-Bench and AgentDojo sources. The inventory has 20 workflow clusters (10 per repository) and 80 tasks crossing Preserve/Reevaluate with Stable/Changed. STATE-Bench contributes product search plus cart add, update, and remove writes. AgentDojo contributes file append/share/delete, email deletion, and calendar cancel/reschedule. Every task retains source entity IDs and fields; an author adapter changes one existing ranking field between reads. The old target remains present and action-valid. Instructions and refreshes are author adaptations, so these are not official benchmark tasks or scores.

Two models run ordinary full history and a matched execution-record probe, yielding 320 rows. The frozen zero-API gate passes 80/80 source-tool final-state checks. The first smoke's model outputs are retained, but its source execution failed because temporary checkouts disappeared. Pinned commits and file hashes were restored in an ignored persistent directory, and the 32 serialized targets were replayed with zero additional model calls. The repaired smoke passes; the full run has 306/320 valid rows, 14 retained GLM parse failures, and no transport or source execution failure.

Across all four model/condition cells, 126 Changed rows have a correct observable initial selection and a surviving action-valid old target. Exact-target success is 118/126. Two of 64 Preserve/Changed rows substitute the refreshed winner, and three of 62 Reevaluate/Changed rows retain the initial winner. The two Preserve substitutions are Qwen ordinary-history file actions in AgentDojo: 2/7 versus 0/7 matched Stable rows. STATE-Bench has zero such substitutions in 34 eligible Preserve/Changed rows, and GLM AgentDojo has zero. Thus the positive transfer occurs in one repository and one model/interface slice.

The execution record does not provide a stable method gain. On shared correct-initial-selection rows, ordinary history versus execution record scores 57/59 versus 52/59 for GLM and 62/65 versus 63/65 for Qwen. Cluster bootstrap estimates execution-record minus ordinary exact-target accuracy at -8.75 points (95% CI $[-16.25, -2.5]$) for GLM and -1.25 points ($[-6.25, 3.75]$) for Qwen. This result supplies limited source-anchored bridge evidence for the diagnostic while narrowing any claim that an explicit record is generally superior. It does not establish native benchmark prevalence, natural traffic, independent human evidence, or an official STATE-Bench or AgentDojo score.

6.5 Custom AppWorld database case study

The custom AppWorld study uses two application/selector clusters. The Todoist cluster was frozen first and uses native `create_task`, `show_tasks`, and `update_task` APIs. A post-primary Simple Note extension uses native `create_note`, `search_notes`, and `add_content_to_note`. Each cluster crosses Preserve/Reevaluate, Stable/Flip, and two paraphrases. Todoist selects the earliest-due incomplete task; Stable adds a later task, and Flip adds an earlier one. Simple Note selects the alphabetically first tagged title; Stable adds `DELTA`, and Flip adds `ALPHA`. Old targets remain present and editable. Gold is computed from stable IDs, and evaluators compare post-sync and final database snapshots without an LLM judge.

App	Agent	Strict	Write	Bind	Subst.	Wrong
Todoist	Qwen3.5	6/8	8/8	6/8	0/6	0
Todoist	GLM-5.1	8/8	8/8	8/8	0/8	0
Simple Note	Qwen3.5	6/8	7/8	6/8	0/6	1
Simple Note	GLM-5.1	4/8	8/8	4/8	0/4	0

Table 38: Custom AppWorld boundary study. Across 24 correct, correctly timed bindings, conditional substitution is 0/24. The one Simple Note wrong write follows synchronization before binding and is excluded from that denominator. Explicit sidecar omission causes the remaining strict failures. This is not an AppWorld leaderboard result; each app contributes one selector cluster.

A post-primary lower-intervention addendum removes `record_binding` and all prompt language about TRI, binding, or temporal authorization. Each ordinary selector API returns one stable ID, which is logged automatically as the binding event. Across 32 Qwen/GLM trajectories, strict success is 28/32, gold-target writes are 30/32, and conditional substitution is 0/28 (0/6 in Preserve/Flip). Qwen's two wrong writes use the same Preserve wording in both apps: it synchronizes before its first selector call, then writes the refreshed winner. Matched Stable rows exhibit the same order defect but happen to write A because the winner is unchanged. This is a reproducible pre-binding temporal-order failure and illustrates why Stable controls and a conditional-substitution denominator are both necessary.

7 Negative Compositional Stress Tests

The frozen Qwen two-refresh multi-referent stress test contains 40 SQLite tasks. Each includes two refreshes, a monitoring referent, an unrelated tool call, and a final mutation. Generic obtains 32/40 final states, one wrong write, and four unnecessary rejections. Scalar Lifecycle obtains 28/40, six wrong writes, six unnecessary rejections, and one parse/internal error. The difference is -10.0 points with a template-cluster interval of $[-35.0, 12.5]$. Lifecycle mode accuracy is only 25/40: its scalar record confuses monitoring and action-target roles.

A frozen role-indexed held-out addendum then tests 40 tasks from four unseen schemas. In the matched scalar-versus-role comparison, Qwen improves from 35/40 to 39/40 final states (difference +10.0, cluster CI [+2.5,+20.0], McNemar $p = .21875$). GLM intention-to-treat is 40/40 scalar versus 37/40 role-indexed because three role runs fail in transport; a serial recovery sensitivity gives 40/40 for both. Against historical CTA, Qwen obtains 38/40 versus 39/40 final states (33/40 vs. 39/40 target accuracy), and recovered GLM ties at 40/40. These results motivate role indexing but do not support stable superiority over simpler compilation.

8 Post-Primary Method-Upgrade Go/No-Go

After the scalar and compositional results above were known, we froze a 20-task decision set: 16 Cross-Schema Controlled Replication tasks over 16 state clusters and four multi-refresh/role tasks. We reused the matching frozen Generic, Historical CTA, Lifecycle, and role-indexed outputs and newly ran an Event Graph compiler (M1) and Event Graph plus Executable Selector (M2) with both models. This is an exploratory method-selection smoke, not a powered confirmatory comparison.

On the same combined tasks, Historical CTA obtains 13/20 for Qwen and 20/20 for GLM. M2 therefore changes accuracy by +2 and -2 tasks, respectively, and fails the predeclared requirements that both models

Model	Method	Correct	Schema	Mode	Selector
Qwen3.5	Event Graph	9/20	14/20	12/20	–
Qwen3.5	Executable Selector	15/20	15/20	14/20	15/20
GLM-5.1	Event Graph	20/20	20/20	20/20	–
GLM-5.1	Executable Selector	18/20	18/20	18/20	19/20

Table 39: Post-primary 20-task method-upgrade smoke. Selector reports equivalence on both initial and final states. Parse/schema failures count as incorrect under intention-to-treat.

reach at least 95% schema/selector equivalence and show a consistent direction. We do not promote M1 or M2 to the main method. Historical CTA remains the minimal scalar realization. The independently frozen role-indexing evidence above motivates role indexing as a limited compositional extension. The zero-API oracle executor is 440/440 across the primary, cross-schema, and compositional inventories and blocks all stale/wrong writes in 120 deterministic concurrency sequences; this establishes implementation sufficiency only, not learned compiler performance.

Leave-group-out sensitivity. On the 240-task Cross-Schema Controlled Replication, we recompute the matched CTA-minus-Generic difference after excluding one complete domain or one complete template family. The ten domain exclusions retain ranges of 18.5–25.5, 23.1–25.9, and 15.7–22.2 points for Qwen, GLM, and DeepSeek; the sixteen template-family exclusions retain 20.4–26.3, 20.8–26.1, and 14.2–19.6 points. Every exclusion remains positive. This rules out dominance by one domain or template family, but does not make templated rows independent natural-world observations.

Within-inventory subsampling stability audit. We subsample complete matched clusters without replacement 10,000 times at increasing cluster counts. This is a precision and stability diagnostic, not post-hoc power. On Matched Timing Diagnostic, the audit reproduces the full Lifecycle-Gated–Generic point differences of +33.8 points for Qwen and +28.1 for GLM; the pre-specified seed-20260717 intervals are reported in the statistical section above. At only 10 of 20 template clusters (80 tasks), 100% of subsamples retain a positive difference for both models. On Cross-Schema Controlled Replication, the full CTA–Generic differences are +23.3 [16.2,30.4], +24.2 [19.6,28.7], and +17.5 [10.8,23.3] for Qwen, GLM, and DeepSeek. At 10 of 40 state clusters (60 tasks), positive differences occur in 100.0%, 100.0%, and 99.8% of draws; at 20 clusters all three reach 100%. Thus the direction is stable under cluster subsampling within the observed inventory. This does not increase the number of natural-world opportunities or justify a prevalence claim.

Temperature-zero repeat stability. Before the repeat calls, we froze one balanced task from each of the 40 Cross-Schema Controlled Replication state clusters. The matching subset of the original run is repeat 1; two new complete passes give three measurements for Qwen and GLM Generic and CTA. CTA–Generic remains positive in every model-repeat cell: Qwen differences are +25.0, +12.5, and +22.5 points, while GLM differences are +30.0, +20.0, and +25.0. Generic conditional substitution is 5/10, 5/11, and 7/11 for Qwen and 4/12, 5/12, and 6/12 for GLM; CTA is zero in every denominator. Exact target unanimity across all three passes is 70.0/72.5% for Qwen Generic/CTA and 85.0/97.5% for GLM. Under the frozen rule this is a Mixed result: the method direction and conditional mechanism are stable, but individual outputs are not fully deterministic. The 320 new task-controller executions use 640 requests, zero retries, and 313,692 provider-reported tokens, with zero API/parse errors.

9 Request Cost

On the 160-task primary set, Generic uses 320 requests per model. Lifecycle-Gated uses 80 Preserve compiler calls and 160 Reevaluate compiler/actor calls, totaling 240. Mean client-observed Qwen latency is 2.84 seconds for gated Preserve versus 3.63 for Generic anchored tasks; corresponding GLM means are 3.47 versus 5.18 seconds. On the 40-task SQLite set, Generic uses 80 requests and Gated uses 60. Token usage was not captured and latency is network-dependent, so these are request-efficiency observations rather than a controlled systems benchmark.

For the post-primary DeepSeek Cross-Schema Controlled Replication run, both Generic and CTA use 480 requests. Provider-reported usage is 264,359 tokens for Generic and 216,446 for CTA; client-observed cumulative latency is 1,651.7 and 1,355.8 seconds. These matched-endpoint observations are reported for provenance, not as a cross-provider systems claim.

10 Blinded Human Agreement Audit

One volunteer uninvolved in TRI design produced natural-language rewrites of 50 sampled scalar instructions. The source script uses seed 20260717 and stratifies the 160-task inventory by the 20 style–update cells, selecting two or three tasks from every cell for a fixed total of 50. Three other volunteers independently labeled 100 differently randomized original/rewrite items. Forms used opaque IDs and concealed source task, binding mode, update, variant, and gold. Responses were an entity ID, Reject, or Clarify; confidence was optional. All three files contain exactly the 100 keyed IDs with no missing, duplicate, invalid, or payload-mismatched rows. All four participants were adults and provided informed consent. The collection instructions described voluntary participation and discouraged names or unnecessary personal data. Two participants received modest one-time honoraria and two were unpaid; compensation was arranged individually and was not contingent on response content or agreement with benchmark gold. The instructions required sufficient English to complete the English-language task, but language background was not independently tested or retained. Recruitment channel, prior relationship to the research team, exact honorarium amounts, and actual completion times were not recorded in the research dataset; the administration guide estimated 45–75 minutes for rewriting and 60–90 minutes per annotator. This convenience-sample limitation precludes demographic generalization. The study was reviewed by the authors' institution and determined to be exempt. No sensitive personal data were collected, and the anonymous Code and Data Supplement contains only de-identified responses; private consent and study-administration records are not included.

Slice	n	Maj.–gold	Unanimous	κ	α
All	100	86.0	72.0	.708	.709
Original	50	84.0	70.0	.679	.681
Human rewrite	50	88.0	74.0	.737	.738
Explicit	60	86.7	70.0	.672	.674
Implicit	40	85.0	75.0	.758	.760
Dynamic	50	98.0	96.0	.924	.925
Anchored actionable	30	86.7	63.3	.538	.543
Anchored reject	20	55.0	25.0	.087	.102

Table 40: Human agreement (percent except coefficients). Majority–gold uses all items, treating six three-way ties as not matching gold. Among 94 determinate majorities, accuracy is 91.5%.

Pairwise raw agreement is 79.3%; 4% of items receive at least one Clarify. All 72 unanimous items match gold. Annotator-level gold agreement is 87%, 80%, and 89%; confidence was absent for annotator 1 and therefore is not used as an outcome. The rewrite slice does not degrade agreement, and explicit/implicit slices are similar. The main weakness is execution fallback: participants often preserve an invalid entity as the discourse referent while disagreeing whether the action answer should be Reject. We therefore interpret the study as descriptive agreement with the Preserve/Reevaluate core, not as evidence for reject or conditional policy. Complete disagreement rows and the auditable analysis script accompany the Code and Data Supplement.

Six-form follow-up audit (failed frozen gate; descriptive boundary). We separately froze a follow-up with six disjoint 12-item forms, five valid labels per item, separate referent and execution judgments, and one reserve slot per form. The cutoff contains 31 complete submissions. All respondents affirmed adulthood, independent English reading, consent, and completion of all 42 response questions. The frozen exclusion rules also required respondents to report no generative-AI, search, machine-translation, or other-person assistance and no technical issue affecting comprehension. Only 11 submissions pass both requirements.

Form	Submitted	Eligibility-passing	Additional valid needed
A	5	1	4
B	5	2	3
C	5	1	4
D	5	1	4
E	6	3	2
F	5	3	2
Total	31	11	19

Table 41: Six-form follow-up at the fixed cutoff. The target is five valid labels in every form, not 30 pooled labels. The failed gate makes this a recruitment and construct-boundary audit rather than a fixed-rater agreement endpoint.

Fifteen respondents reported prohibited assistance and 11 reported a comprehension-affecting technical issue; six reported both, giving nine assistance-only, five technical-only, and six joint exclusions. The 11 eligible submissions provide 132 labels, distributed as 36 items with one label, 12 with two, and 24 with three. Therefore no item has the required five valid labels, and the frozen Fleiss' κ, Krippendorff's α, and 18-pair endpoint are unavailable. For diagnosis only, eligible label-level agreement with author gold is 51/132 (38.6%) for referent identity and 34/132 (25.8%) for execution. The median completion time is 330 seconds, and 30/31 submissions finish below ten minutes versus the planned 12–15 minutes.

We retain two non-evidentiary sensitivities rather than discarding the excluded responses. A post-hoc slice that ignores the assistance exclusion but retains the technical-issue exclusion has 20 participants, with referent agreement 92/240 (38.3%) and execution agreement 61/240 (25.4%). The assistance question combines translation, generative AI, search, and another person's help, so this slice cannot isolate benign translation. The widest sensitivity uses all 30 primary submissions: referent majority agrees with gold on 27/72 items, execution majority on 19/72, and referent changed-pair accuracy is 3/18. These outcomes do not rescue the audit and cannot be compared as a completed replication of the earlier three-annotator study.

We classify this completed cutoff as a post-primary failed-gate audit. It supplies descriptive boundary evidence and does not strengthen the TRI gold. At least 19 additional eligibility-passing responses are needed, distributed 4/3/4/4/2/2 across A–F. Because the frozen allocation included only one reserve per form, further recruitment requires a prospective amendment before collection. A repaired study should keep the frozen items and gold, separate translation from AI/search/other-person help, add a pre-item comprehension gate, and recruit from a monitored panel. The private platform exports and response mapping are excluded from the anonymous artifact. The public aggregate outputs retain only form-level source hashes, de-identified eligibility codes, and per-item response counts.

Human-majority model sensitivity. Of the 50 original items, 46 have a determinate human majority, 42 majorities match benchmark gold, and 35 are unanimously gold-aligned. Re-scoring frozen model outputs against the human majority gives:

Model	Generic	Lifecycle-Gated	Historical CTA
Qwen3.5	32/46 (69.6)	41/46 (89.1)	42/46 (91.3)
GLM-5.1	34/46 (73.9)	42/46 (91.3)	41/46 (89.1)

Table 42: Accuracy against determinate human-majority targets (percent in parentheses).

Gated-minus-Generic template-cluster gaps are +19.6 points with a 95% interval of [+5.0,+36.4] for Qwen and +17.4 [+6.8,+31.6] for GLM (18 clusters, 10,000 draws). Secondary task-level exact McNemar tests give $p = .0117$ and .0078. On the 42 majority-gold items, Gated reaches 97.6% and 100%; on the 35 unanimous-gold items it reaches 97.1% and 100%. This sensitivity preserves the Generic-to-compiled gap but does not show Lifecycle-Gated superiority over historical pre-refresh compilation.

11 Independent Human-Rewrite Model Evaluation

After the human study and before model calls, we froze 50 instructions rewritten by one independent volunteer and reran the unchanged four controllers. All eight files contain exactly 50 unique expected IDs, all 400 rows have `status=ok`, and no request retry occurred.

Model	Generic	CTA	Lifecycle-free	Gated
Qwen3.5, benchmark gold	60.0	90.0	88.0	90.0
GLM-5.1, benchmark gold	74.0	98.0	92.0	94.0
Qwen3.5, human majority	60.4	89.6	83.3	85.4
GLM-5.1, human majority	68.8	93.8	83.3	85.4
Qwen3.5, human actionable	68.3	92.7	80.5	82.9
GLM-5.1, human actionable	68.3	92.7	80.5	82.9

Table 43: Accuracy (%) on all 50 rewrite golds, the 48 determinate rewrite majorities, and the 41 majority-actionable items (excluding majority-Reject).

On the 37 unanimous-gold rewrites, Generic/CTA/Lifecycle-free/Gated reach 75.7/97.3/86.5/89.2% for Qwen and 81.1/100/89.2/91.9% for GLM. On the 25 dynamic items, CTA/Gated reach 96/80% and 100/88%, respectively; complete explicitness, update, template, error-type, and majority-gold slices are in the artifact. CTA-minus-Generic benchmark-gold differences are +30.0 points with a template-cluster interval of [+11.3,+50.0] for Qwen and +24.0 [+8.3,+41.3] for GLM. Gated reaches the same Qwen gold accuracy as CTA but is four points lower for GLM and lower against human majorities for both models. Direct CTA-minus-Gated intervals are 0.0 $[-11.1, +11.1]$ for Qwen and +4.0 $[-4.3, +14.8]$ for GLM; CTA-minus-Lifecycle-free intervals are +2.0 $[-8.5, +12.2]$ and +6.0 $[-3.6, +16.7]$, respectively. None excludes zero, so the experiment does not establish CTA superiority over either typed variant. All five Qwen and three GLM Gated benchmark-gold errors are dynamic items compiled as Preserve; one is human-contested. CTA's errors are predominantly author-specified invalid-target outcomes. The result therefore supports pre-refresh commitment compilation on this volunteer-authored rewrite set, not Lifecycle-Gated superiority. The execution log discloses two pre-call empty-run failures and the non-balanced health-smoke deviation.

Matched-call transfer. A separate post-primary run applies the full-diagnostic History-only and Decision-visible actors to the same 50 frozen rewrites. The 50 rows form 43 clusters; only seven clusters contain both timing modes, and only three are actionable changed-winner pairs. PairAcc is therefore a small-sample sensitivity rather than the main endpoint.

Model	Outcome	Actionable E2E	Human majority	Changed PairAcc	Preserve subst.
Qwen	History-only	30/40	34/48	0/3	9/10
Qwen	Decision-visible	30/40	35/48	1/3	8/10
Qwen	Decision-enforced	33/40	39/48	2/3	0/10
GLM	History-only	31/40	33/48	1/3	9/10
GLM	Decision-visible	39/40	41/48	3/3	0/10
GLM	Decision-enforced	40/40	43/48	3/3	0/10

Table 44: Matched-call results on one-volunteer rewrites of authored tasks. Human-majority accuracy uses 48 determinate labels; actionable E2E excludes ten author-specified Reject rows.

Decision-visible minus History-only actionable E2E is 0.0 points [−10.0,9.5] for Qwen and +20.0 [8.6,32.5] for GLM. All 300 planned logical calls completed without transport or parse failure. The mixed Qwen result and the three-pair changed subset are retained. This test uses one-volunteer wording of authored semantics and is not an open-language benchmark.

12 Model-Authored Linguistic Stress Audit

This post-primary audit tests dependence on the authors' surface templates without treating model output as independent human language. Before the calls, we froze 24 actionable changed-winner workflow specifications over 24 new schemas. DeepSeek-V4-Pro then wrote one Preserve and one Reevaluate instruction per specification without seeing the original instructions, Rule* cues, controller outputs, or entity IDs. Gold follows only from the frozen event order and states. All 48 generated rows remain in ITT. Qwen3.5 and GLM-5.1 separately classified timing and checked selector, action, and ambiguity; these labels are a model-assisted sensitivity, not independent annotation.

All 24 authoring calls, 96 judge calls, and 384 full controller calls completed without retry or parse failure. Qwen accepted 25/48 instructions and GLM accepted 33/48, but their intersection contains only 11 rows and no complete opposite-gold pair. Qwen classified 23/24 Reevaluate rows as Preserve, while GLM classified 14/24 Preserve rows as Reevaluate. Both marked selector and action fidelity and unambiguity true on all rows. Because no complete pair was accepted by both judges, their labels cannot support a joint PairAcc sensitivity. The disagreement is itself a construct-validity failure for model-only replacement of independent writers and annotators.

Model	Controller	Row acc.	PairAcc	Preserve substitution	Failures
Qwen3.5	Generic	24/48	0/24	24/24	0
Qwen3.5	CTA	36/48	12/24	0/12	0
GLM-5.1	Generic	24/48	0/24	20/24	0
GLM-5.1	CTA	36/48	12/24	0/12	0
Rule*	post-hoc	15/48	5/24	–	28 unresolved

Table 45: Post-primary model-authored stress audit. CTA–Generic PairAcc is +50.0 points with state-pair bootstrap interval $[+29.2, +70.8]$ for each model. Conditional substitution requires an exact correct initial ID. No complete pair passes both model judges, so these results do not establish open-language transfer.

The first aggregate report incorrectly scored every model row as wrong because the new `MAS-01-A`-style IDs contain two hyphens, whereas the pre-existing normalizer retained only `MAS-01`. The invalid all-zero report and raw rows are preserved. Before complete-set reparsing, we froze a zero-request transport repair that accepts only a final actor `target_id` or compiled ID exactly equal to a serialized state ID; it uses no fuzzy, display-name, or semantic matching. The table reports the repaired audit with a post-primary, transport-repaired evidence status. The ITT direction is consistent with the controlled diagnosis, but model authoring and judge disagreement leave the open-language concern unresolved.

13 Code and Data Supplement Map

The separately uploaded anonymous Code and Data Supplement contains the frozen JSONL task files, raw model outputs, SQLite traces, report scripts, controller implementations, protocol files, SHA-256 manifests, and tests. The directory `tri_artifact/reports/` contains the primary cluster reports, component audits, SQLite replay reports, ToolSandbox pilot report, and claim-provenance records, `claims_to_evidence.csv`, matched-pair and qualitative-case reports, coverage checklists, official opportunity/trace audits, and the AppWorld custom protocol. The revision matched-call inventories, append-only execution log, raw outputs, v2 reports, and figure generator are stored under the same `data/`, `runs/`, `reports/`, and `scripts/` directories. Running `PYTHONPATH=. python scripts/audit_main_paper_evidence.py` rebuilds the paper-facing evidence audit from frozen raw outputs and source reports and checks the critical diagnostic-table rows and headline claims against the manuscript source. Current submission runners receive credentials through environment variables; credentials are not embedded in or stored with the artifact.

AI-assisted engineering disclosure. Generative AI systems were used for language editing, code assistance, internal review, and the disclosed model-authored linguistic stress audit. They were not treated as authors, citable sources, independent annotators, or independent human evidence. The authors verified the manuscript claims, references, code, and reported results.

Model and environment provenance. The paper-facing API runs use SiliconFlow's OpenAI-compatible `https://api.siliconflow.cn/v1/chat/completions` endpoint and the exact provider model IDs `Qwen/Qwen3.5-122B-A10B`, `Pro/zai-org/GLM-5.1`, and `deepseek-ai/DeepSeek-V4-Pro`. The provider did not expose immutable weight revisions or serving hardware, which limits exact API replay. Frozen primary and cross-schema calls use temperature zero, thinking disabled, and a 1,200-token output cap. Per-row timestamps, request attempts, retries, latency, and configuration are stored in raw JSONL. Protocol files record timeout, retry, smoke, and stopping rules for each extension, while prompts are literal strings in the versioned runners. Paper-facing primary and replication calls span July 17–26, 2026 (UTC). Offline reports and tests were verified on macOS 26.5.1 arm64 with Python 3.12.13 and pytest 9.1.1; core generation, scoring, bootstrap, and SQLite replay use the Python standard library. These local details reproduce analysis from frozen outputs, not provider inference.

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